



UNIVERSITÀ DEGLI STUDI
DI SALERNO



A Vision Transformer Model for EEG-Based Motor Imagery: enhancing generalization through Self-Supervised Pre-Training

Master Degree in Computer Engineering

Supervisors:

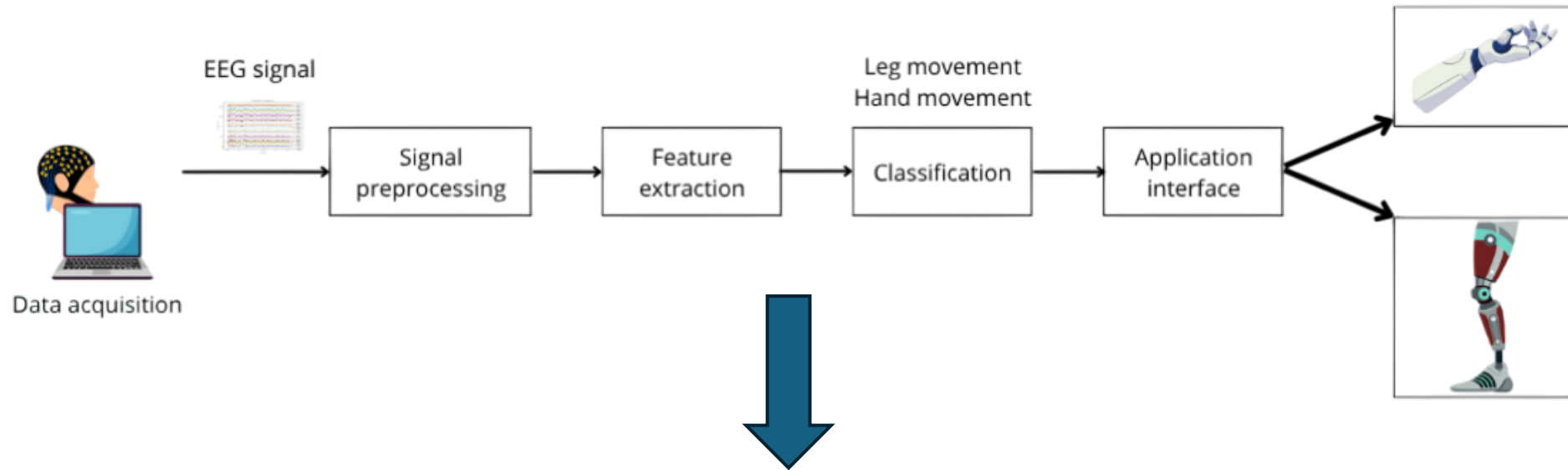
Prof. Martino Giaquinto
Prof.ssa Alessia Saggese
Prof. Mario Vento

Candidate:

Christian Salvatore De Angelis
Serial Number: 0622702349

What is motor imagery?

Motor imagery (MI) refers to the mental rehearsal of a motor action, such as moving a hand or a leg, without its actual execution.

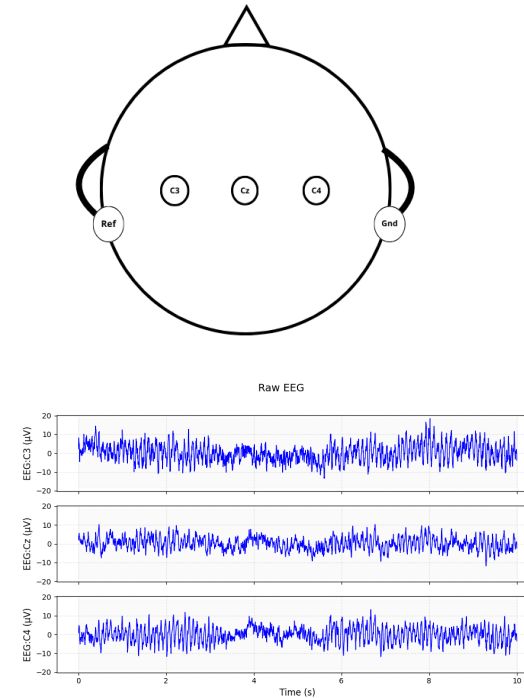
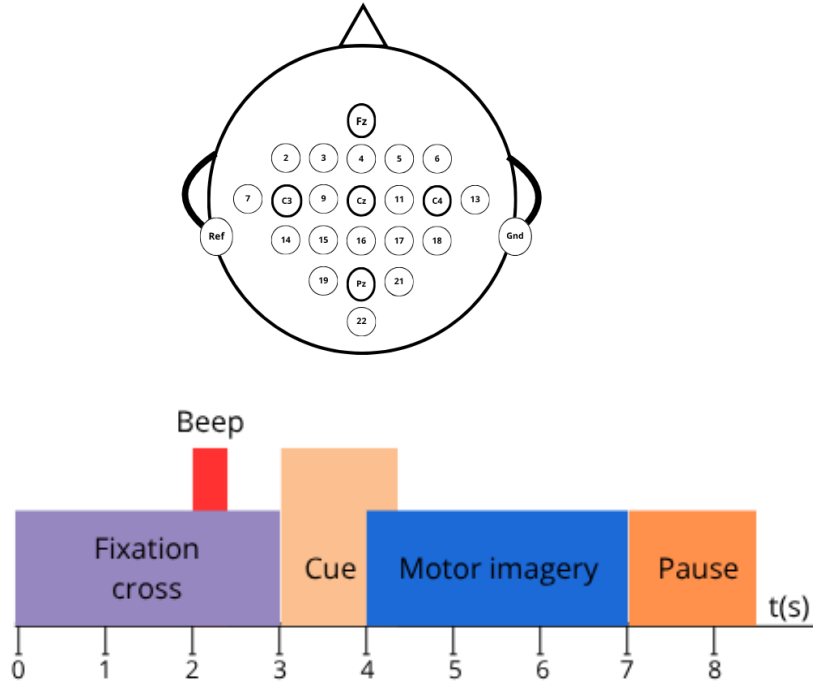
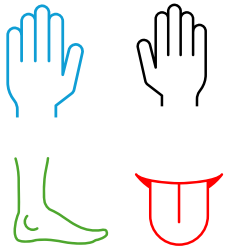


Executed by a Brain Computer Interface (BCI)

Datasets: Bci Competition Iv 2a and 2b

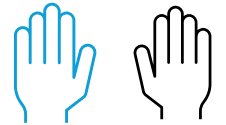
2a [1]

- 9 subjects
- 4 task: left hand,
- right hand, both feet,
- tongue
- 288 trials per session
- 22 electrodes



2b [1]

- 9 subjects
- 2 task: left hand,
- right hand
- 120 trials per session,
- 5 sessions
- 3 electrodes (C3, Cz, C4)



Data scarcity



Recording session are particularly demanding.



Noise

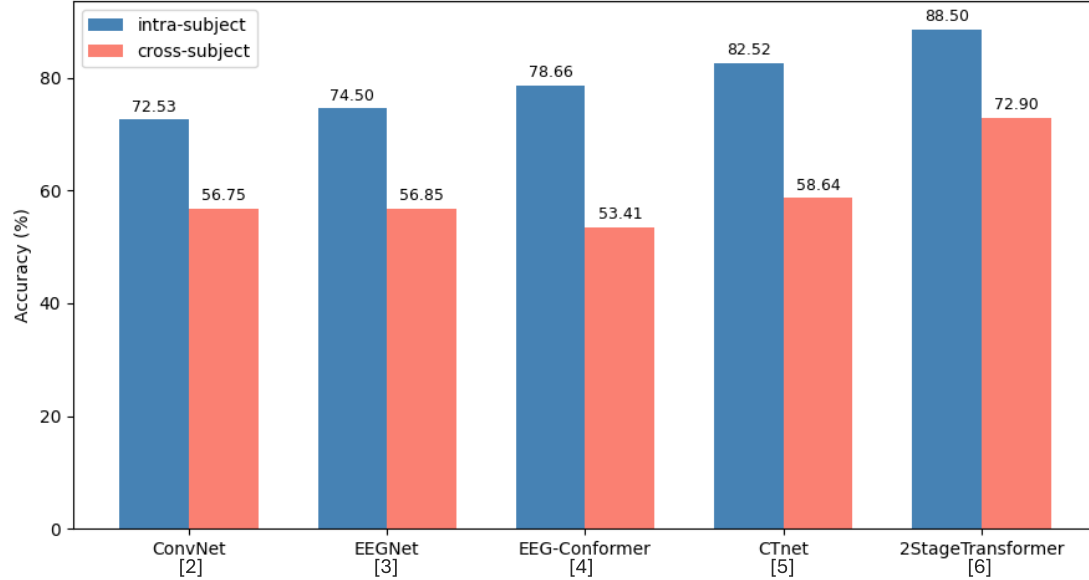


Biological and non-biological noise.

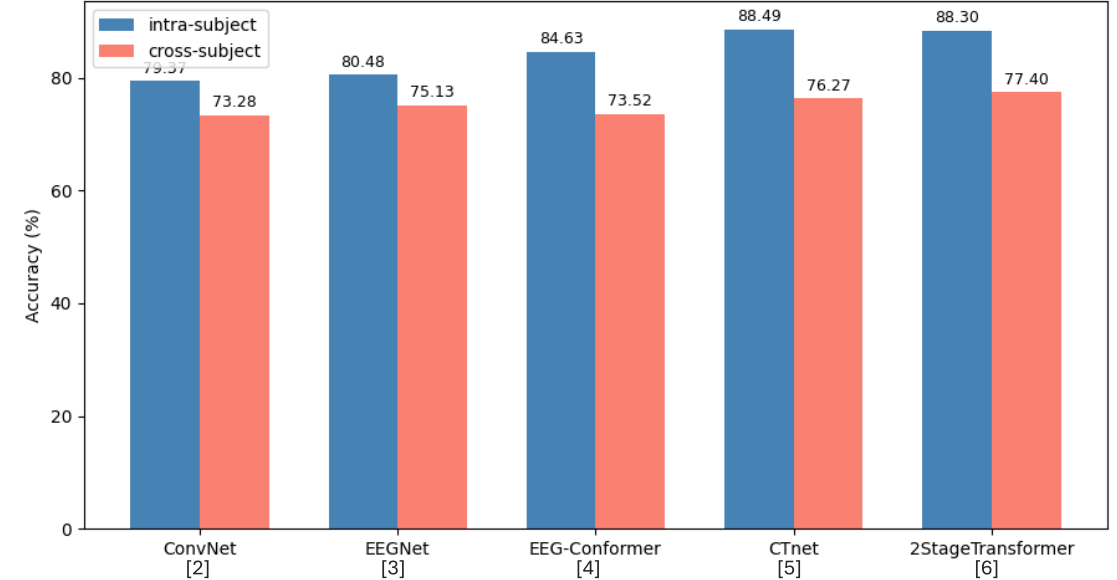


Intra-subject vs cross-subject

Accuracy Comparison Between Intra-subject and Cross-subject on 2a



Accuracy Comparison Between Intra-subject and Cross-subject on 2b



Intra-subject
Cross-subject



**One model per
subject or a single
model for all subjects**



**Non-stationary
EEG data**

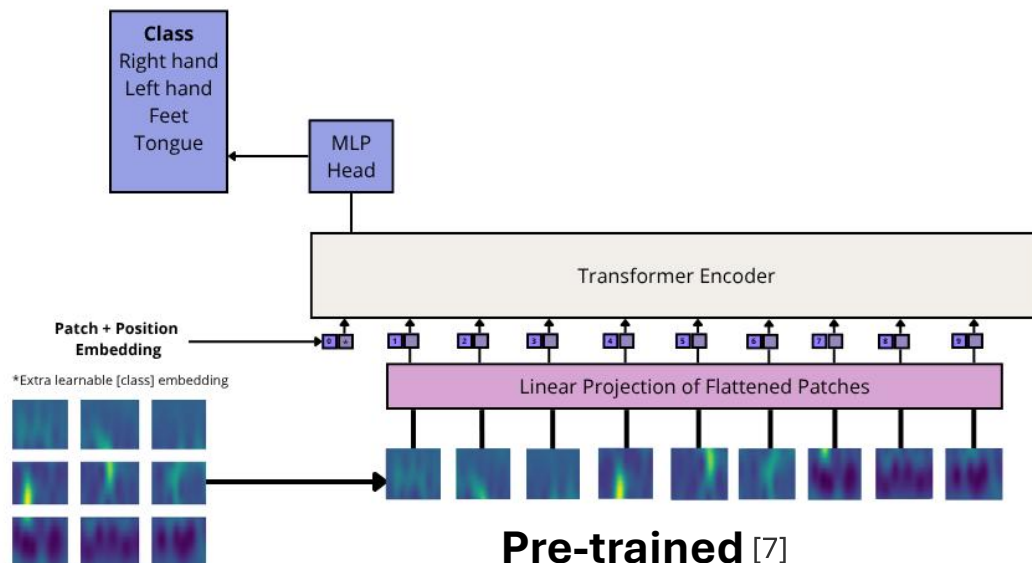


**Different data
distributions
for each subject makes
generalization difficult.**



Personal contribution

- A Multi Channel Vision Transformer model for motor imagery task
Comparison across different ViT architectures
- Proposal of a pre-training framework for the motor imagery task
- Pre-train results analysis



Data
scarcity

Intra-subject
Cross-subject

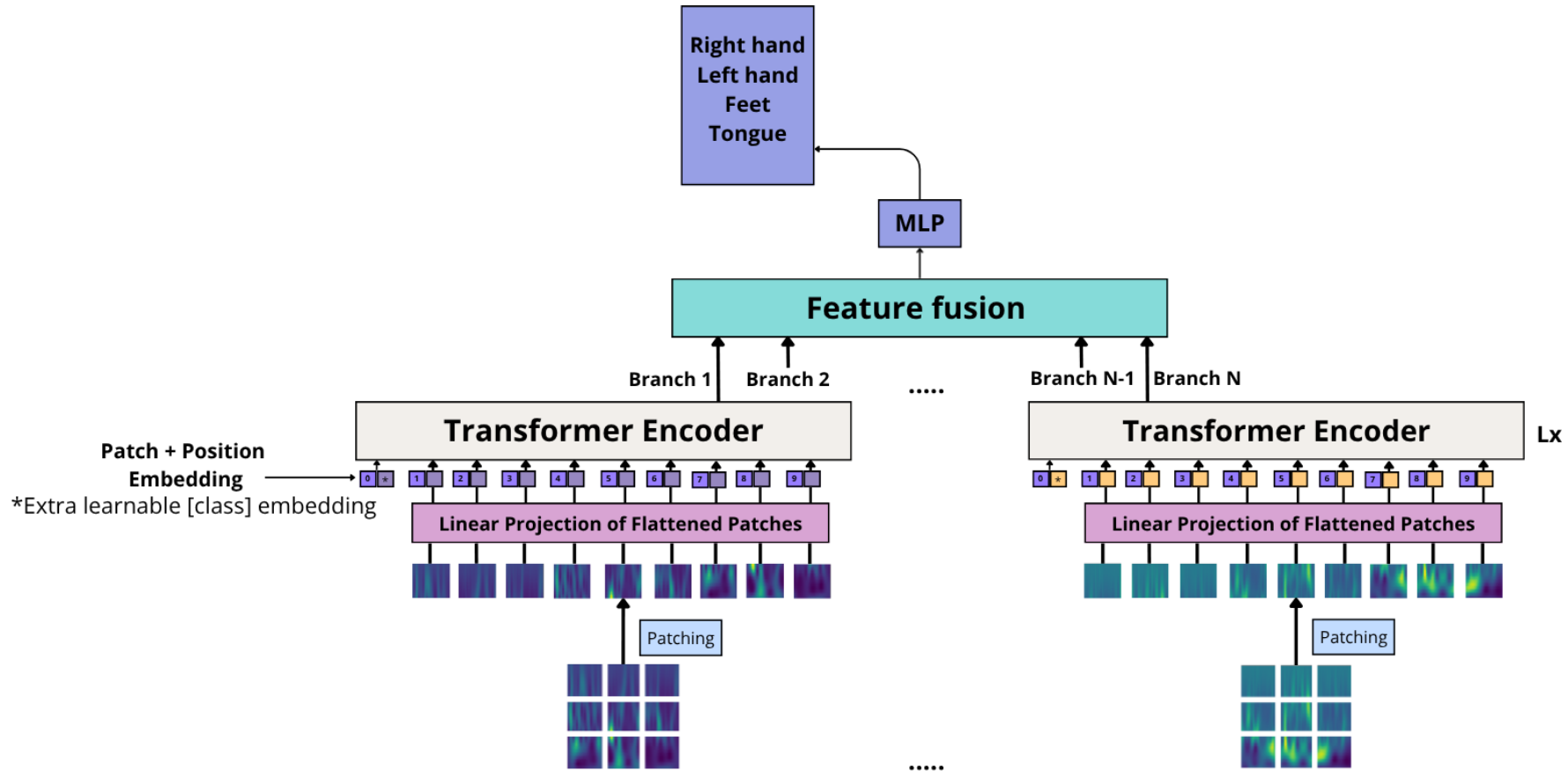


Noise

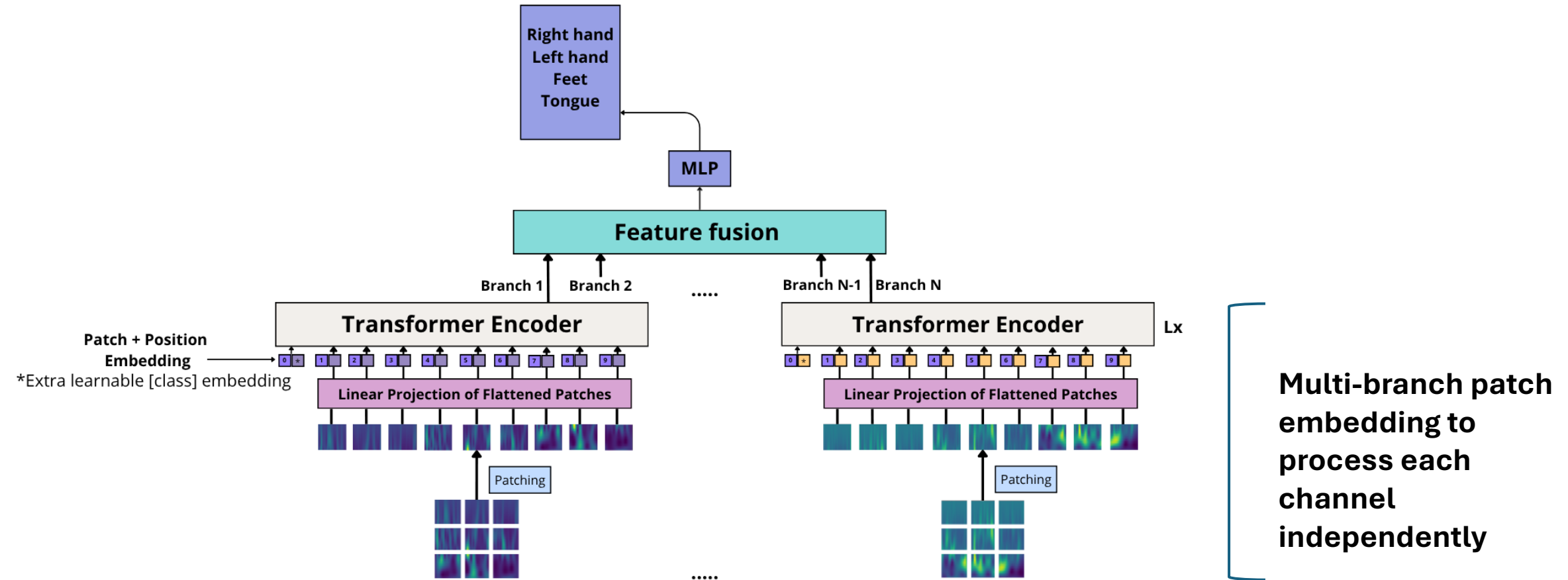
Non-stationary
EEG data



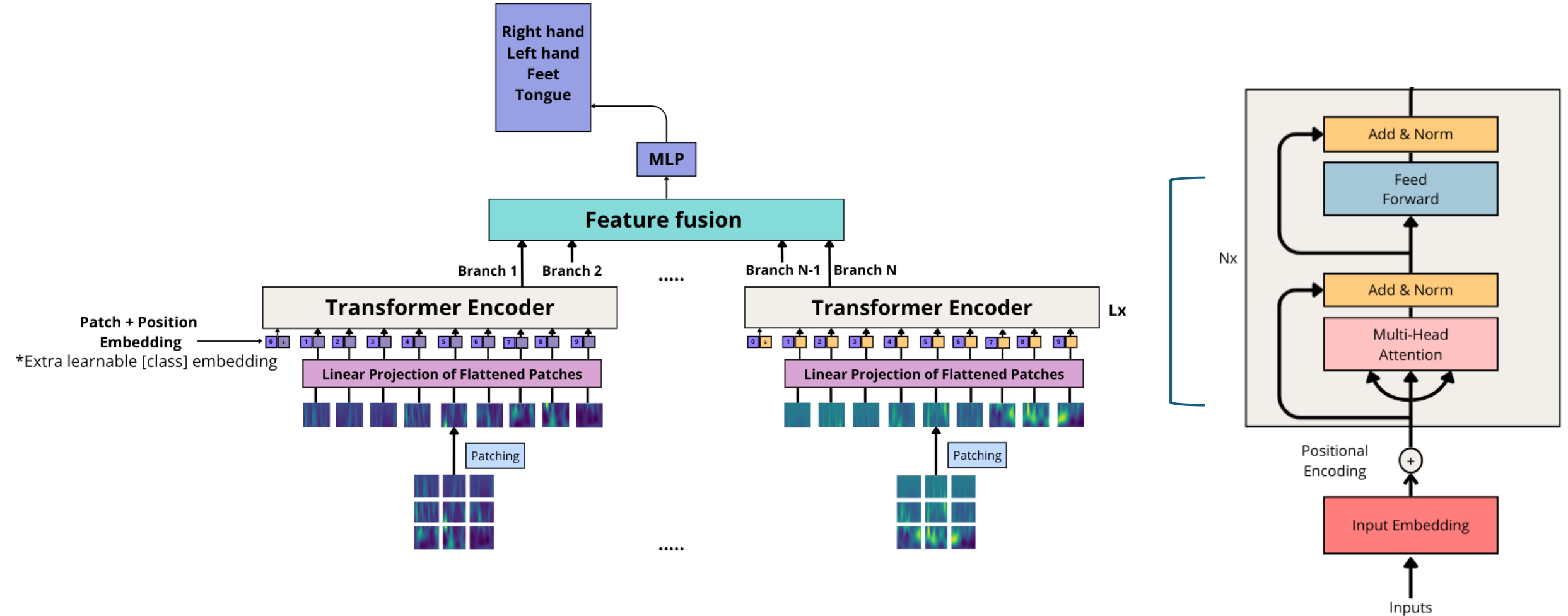
EEG Multi-Channel Vision Transformer



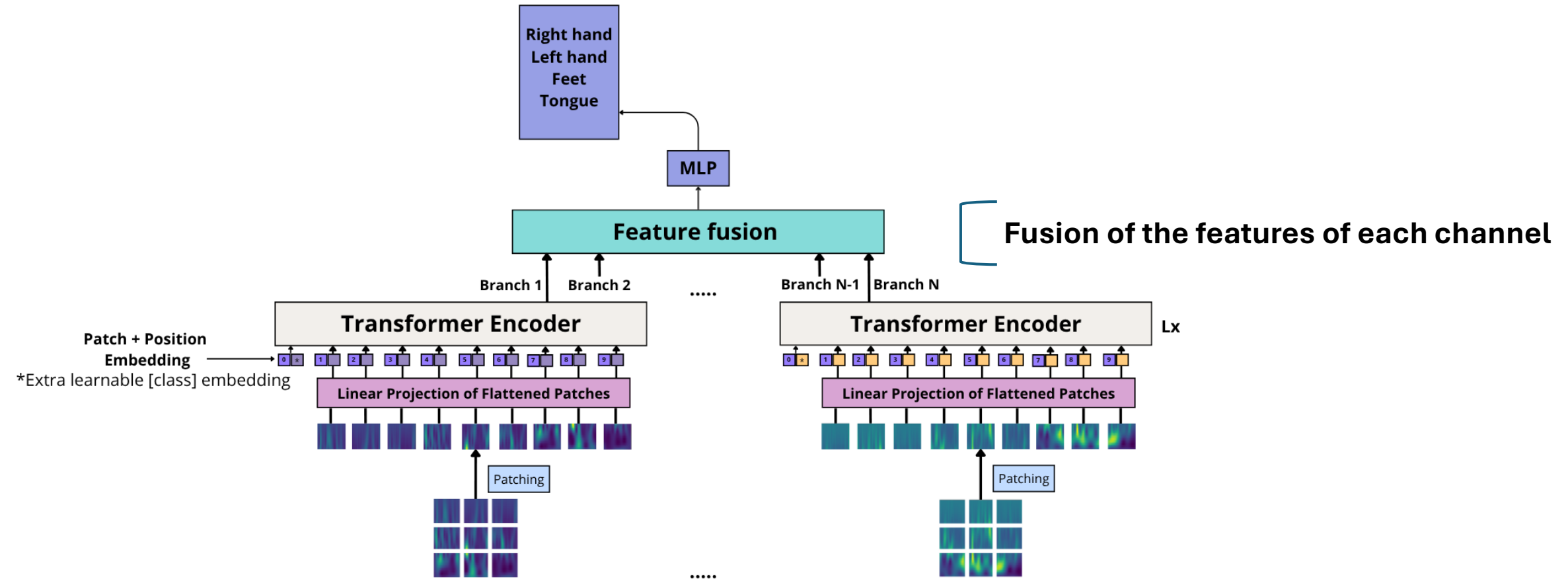
EEG Multi-Channel Vision Transformer



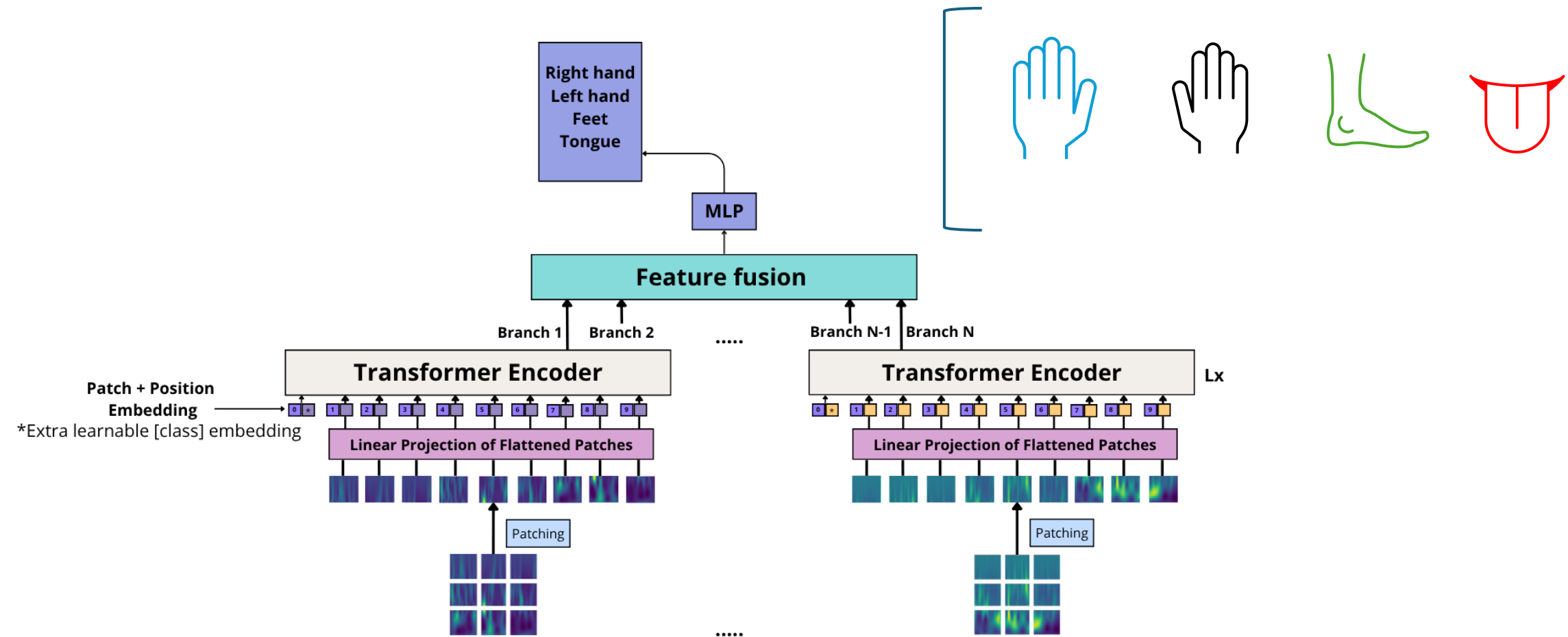
EEG Multi-Channel Vision Transformer



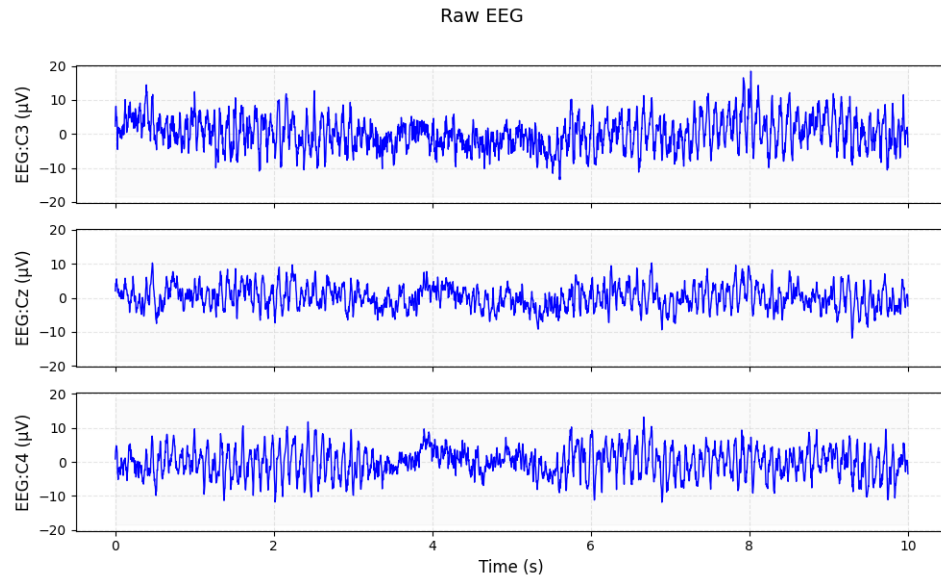
EEG Multi-Channel Vision Transformer



EEG Multi-Channel Vision Transformer

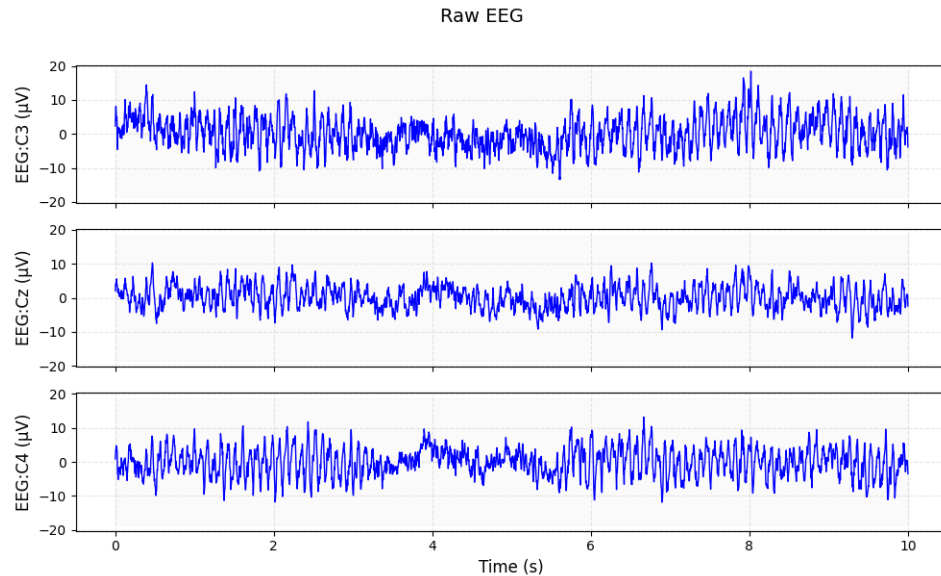


From raw EEG to spectrogram: Wavelet transform



**The raw EEG signal is a 2D array
(channels × time),
while the Vision Transformer
requires a 3D input.**

From raw EEG to spectrogram: Wavelet transform



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(channels × time),
while the Vision Transformer
requires a 3D input.**



Time-frequency transform

Time-frequency transforms

Short time Fourier
transform (STFT)

Mel transform

Continuous Wavelet
transform (CWT)

Time-frequency transforms

Short time Fourier transform (STFT)

- Selection of window type, size and stride.
- Static window.
- Time-frequency trade-off.

Mel transform

- Logarithmic mapping to high frequencies.
- Frequency axis remapped onto the Mel scale.
- Diffused in audio literature, not EEG

Continuous Wavelet transform (CWT)

- Dynamic window for time-frequency trade-off.
- Emphasize alpha, beta and theta bands.
- Diffused in EEG literature.

Time-frequency transforms

Short time Fourier transform (STFT)

- Selection of window type, size and stride.
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Continuous Wavelet transform (CWT)

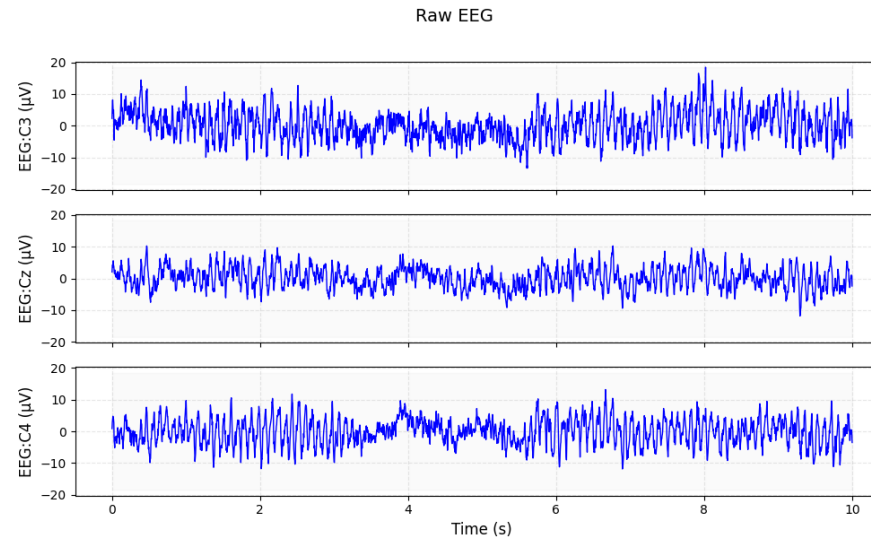
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From raw EEG to spectrogram: Wavelet transform

Continuous Wavelet transform (CWT)

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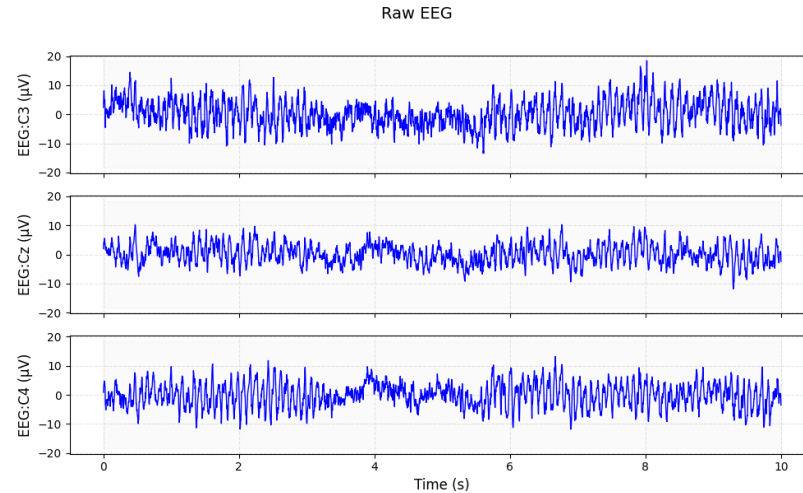
From raw EEG to spectrogram: Wavelet transform

Continuous Wavelet transform (CWT)

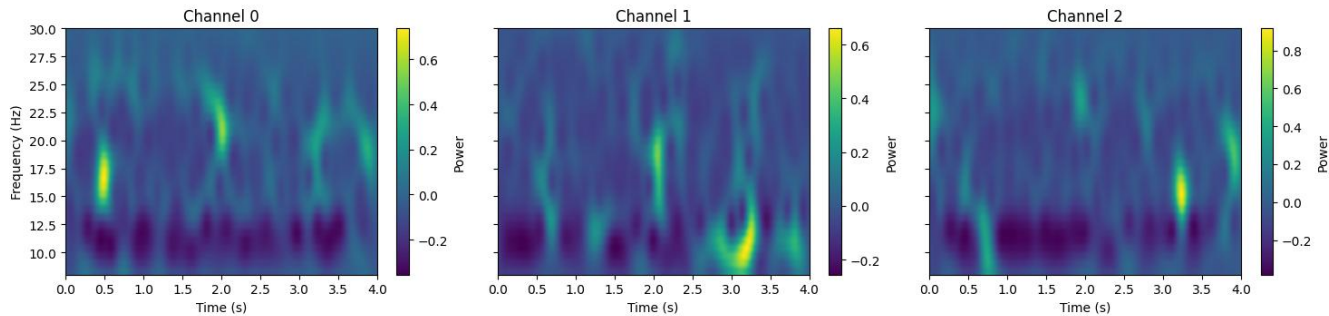
- Dynamic window for time-frequency trade-off.
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- Diffused in EEG literature.



3D data [C × F × T], C channels, F frequency bins, and T time frames.

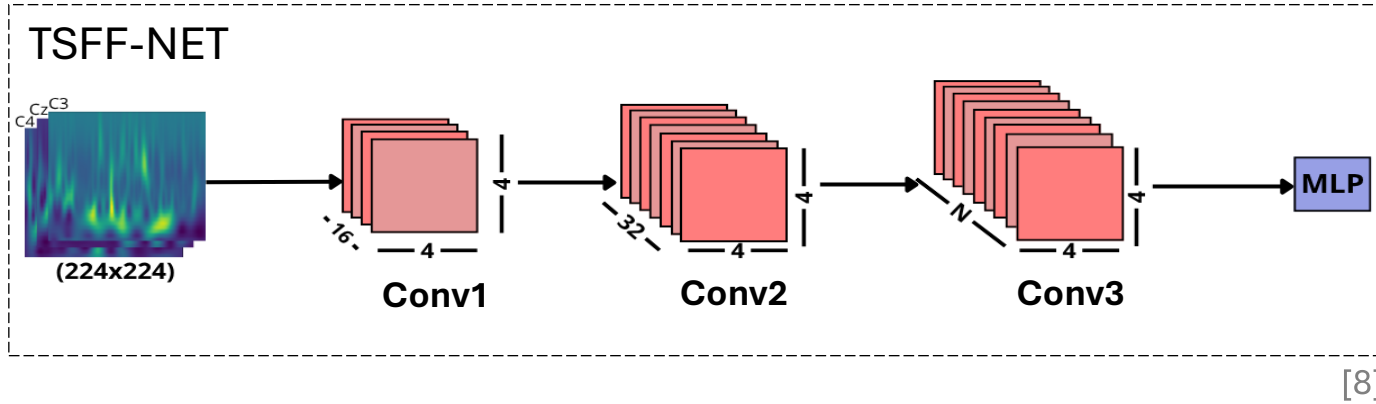


Preprocessing



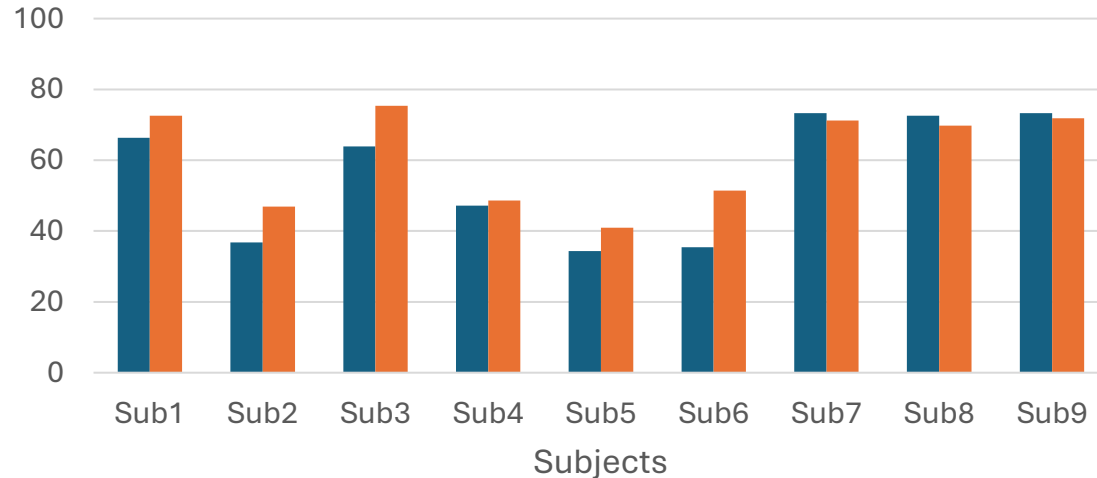
EEG Multi-Channel Vision Transformer

Intra-subject performance

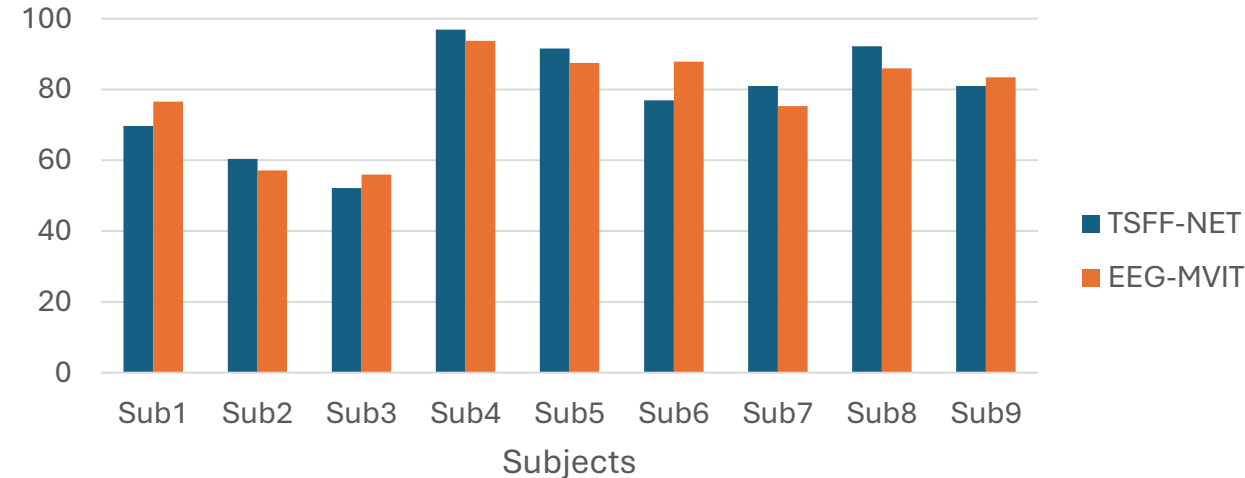


Model	Dataset	MeanAcc	MeanKappa
TSFF-NET	2a	55.90	0.41
EEG-MVIT		61.00	0.48
TSFF-NET	2b	77.96	0.56
EEG-MVIT		78.15	0.56

TSFF-NET vs EEG-MVIT
Bci Competition Iv 2a

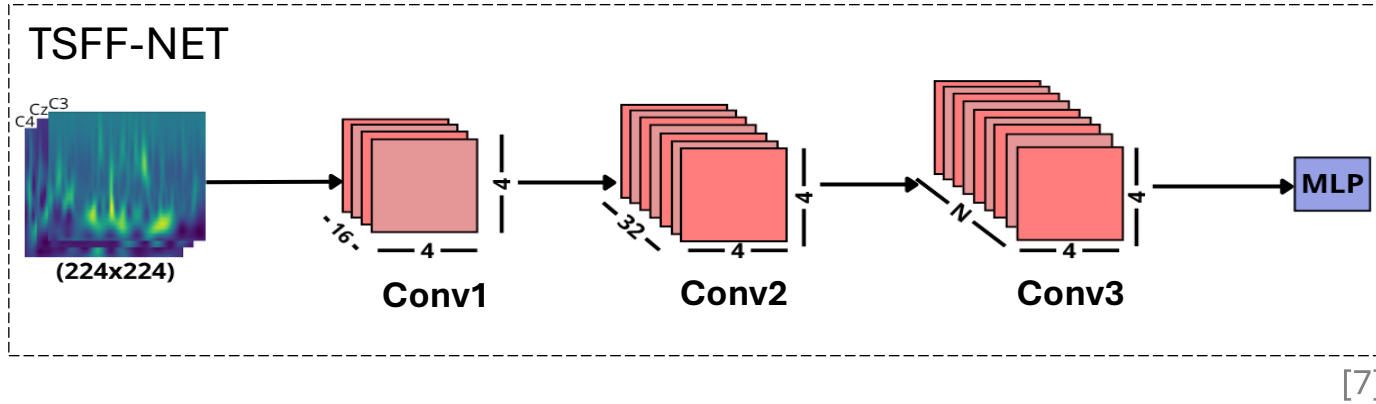


TSFF-NET vs EEG-MVIT
Bci Competition Iv 2b



EEG Multi-Channel Vision Transformer

Intra-subject performance



Model	Dataset	MeanAcc	MeanKappa
TSFF-NET	2a	55.90	0.41
EEG-MVIT		61.00	0.48
TSFF-NET	2b	77.96	0.56
EEG-MVIT		78.15	0.56

Although EEG-MVIT outperforms TSFF-net, its accuracy on 2a is 61% and on 2b is 78.15%, which are not very high values.



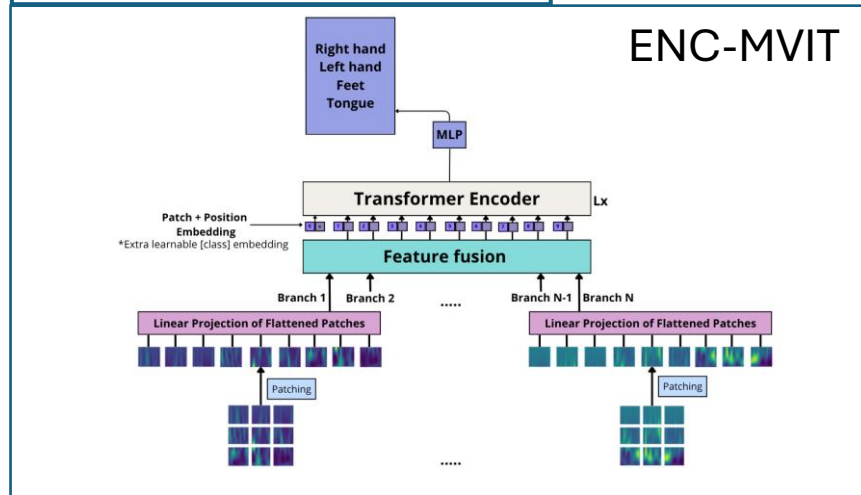
Why?

Surely the attention mechanism offers advantages, but it is necessary to investigate what else impacts performance.

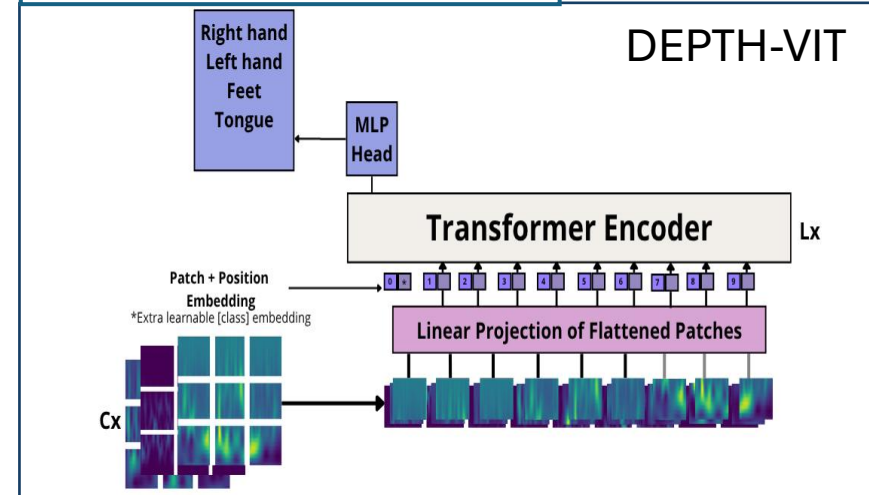
EEG Multi-Channel Vision Transformer

Intra-subject performance

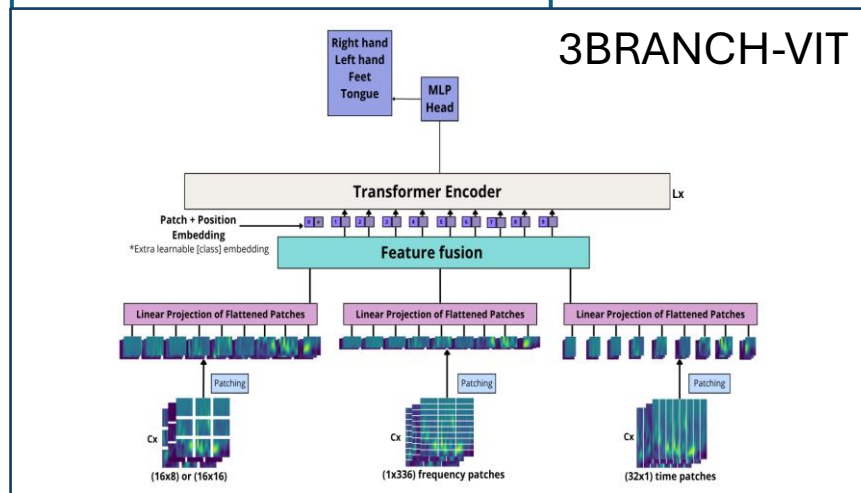
Feature Fusion



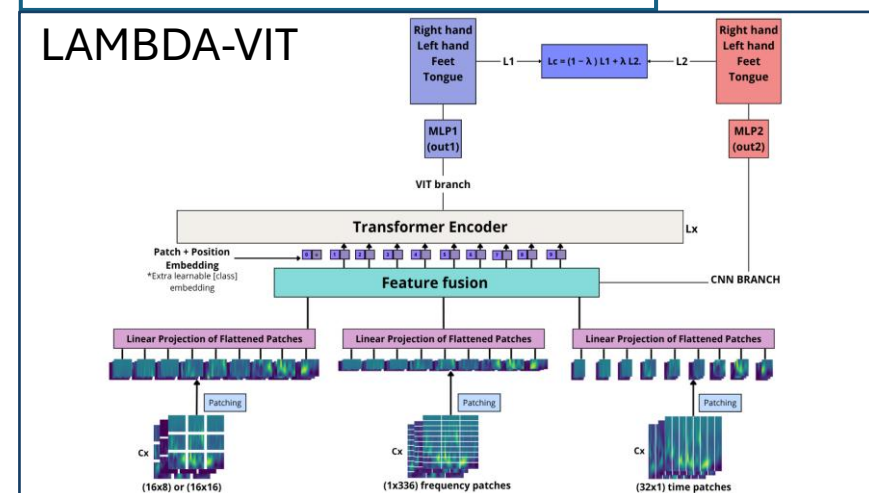
Channel structure



Patch Embedding

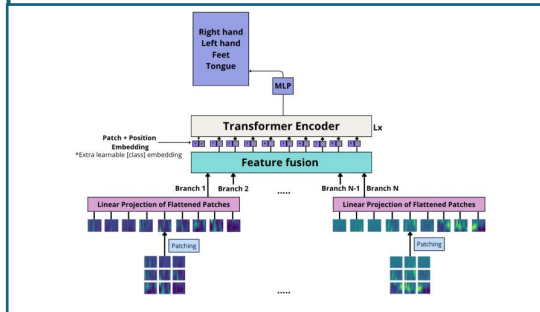


CNN impact



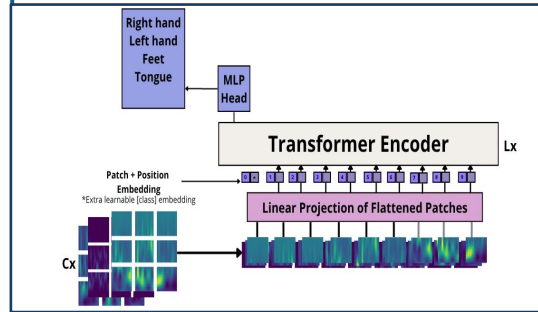
Intra-subject performance on 2a

Feature Fusion



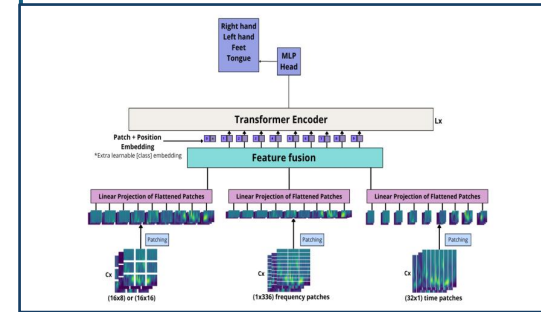
ENC-MVIT

Channel structure



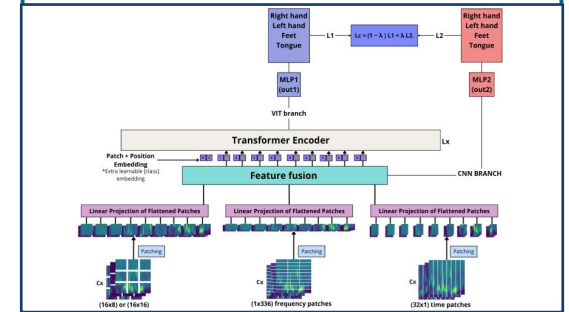
DEPTH-VIT

Patch Embedding



3BRANCH-VIT

CNN impact

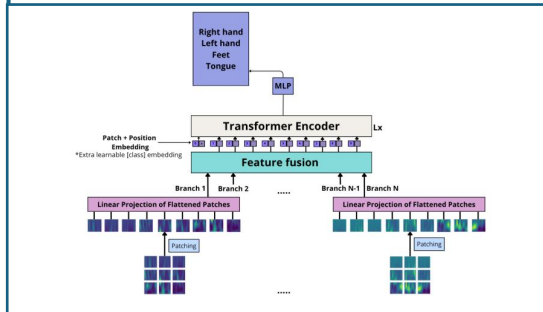


LAMBDA-VIT

Model	MeanAcc±std	MeanKappa±std
EEG-MVIT	61.00±13.63	0.48 ± 0.18
ENC-MVIT	59.91±13.93	0.46 ± 0.18
3BRANCH-VIT	63.39±15.11	0.51 ± 0.20
DEPTH-VIT	65.90±13.06	0.54 ± 0.17
LAMBDA-VIT	66.98±15.17	0.56 ± 0.20

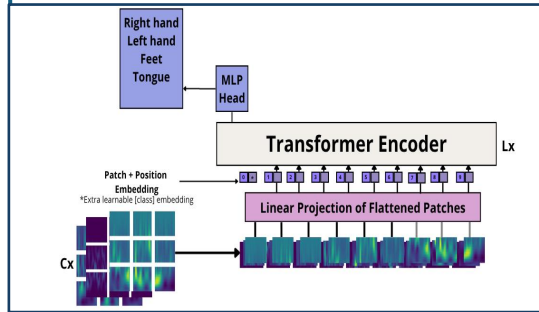
Intra-subject performance on 2b

Feature Fusion



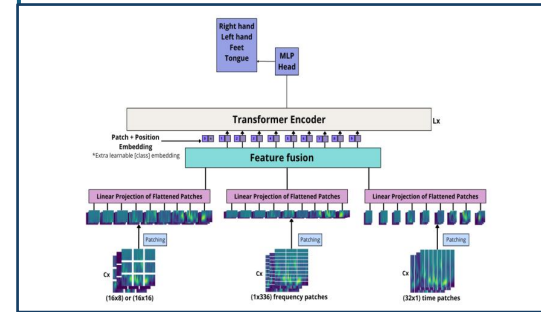
ENC-MVIT

Channel structure



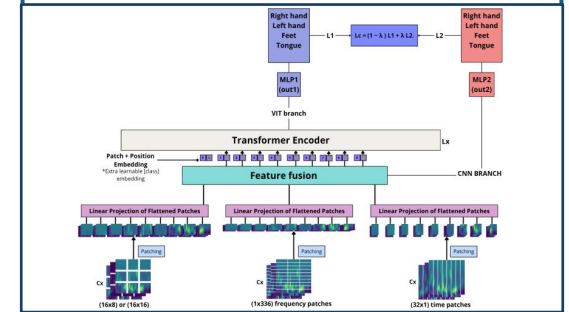
DEPTH-VIT

Patch Embedding



3BRANCH-VIT

CNN impact

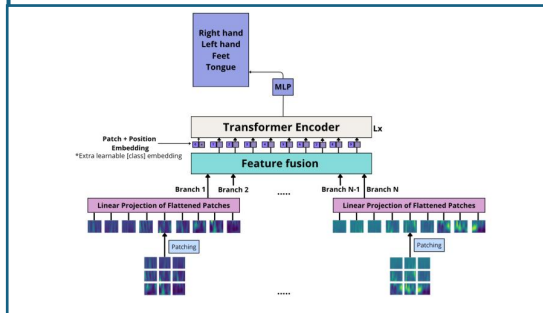


LAMBDA-VIT

Model	MeanAcc±std	MeanKappa±std
EEG-MVIT	78.15±13.50	0.56 ± 0.27
ENC-MVIT	79.12±14.38	0.58 ± 0.28
3BRANCH-VIT	80.00±14.77	0.60 ± 0.29
DEPTH-VIT	78.63±13.69	0.57 ± 0.27
LAMBDA-VIT	80.00±15.30	0.60 ± 0.30

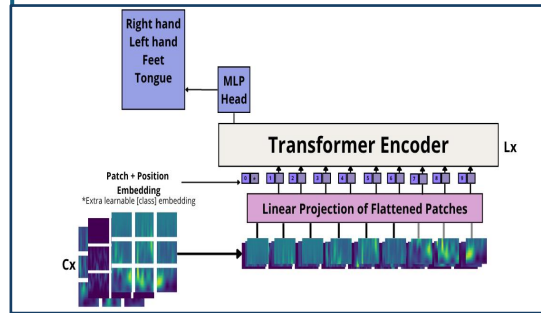
Intra-subject performance on 2b

Feature Fusion



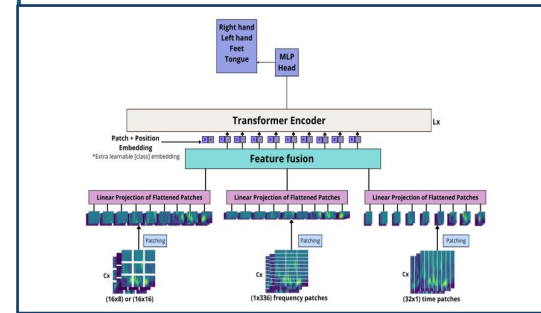
ENC-MVIT

Channel structure



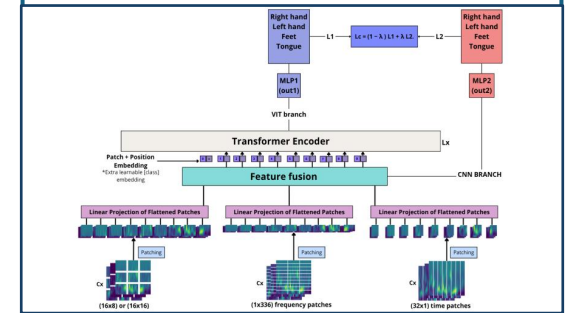
DEPTH-VIT

Patch Embedding



3BRANCH-VIT

CNN impact



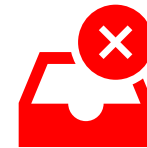
LAMBDA-VIT

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DEPTH-VIT	78.63±13.69	0.57 ± 0.27
LAMBDA-VIT	80.00±15.30	0.60 ± 0.30

The CNN effect is important

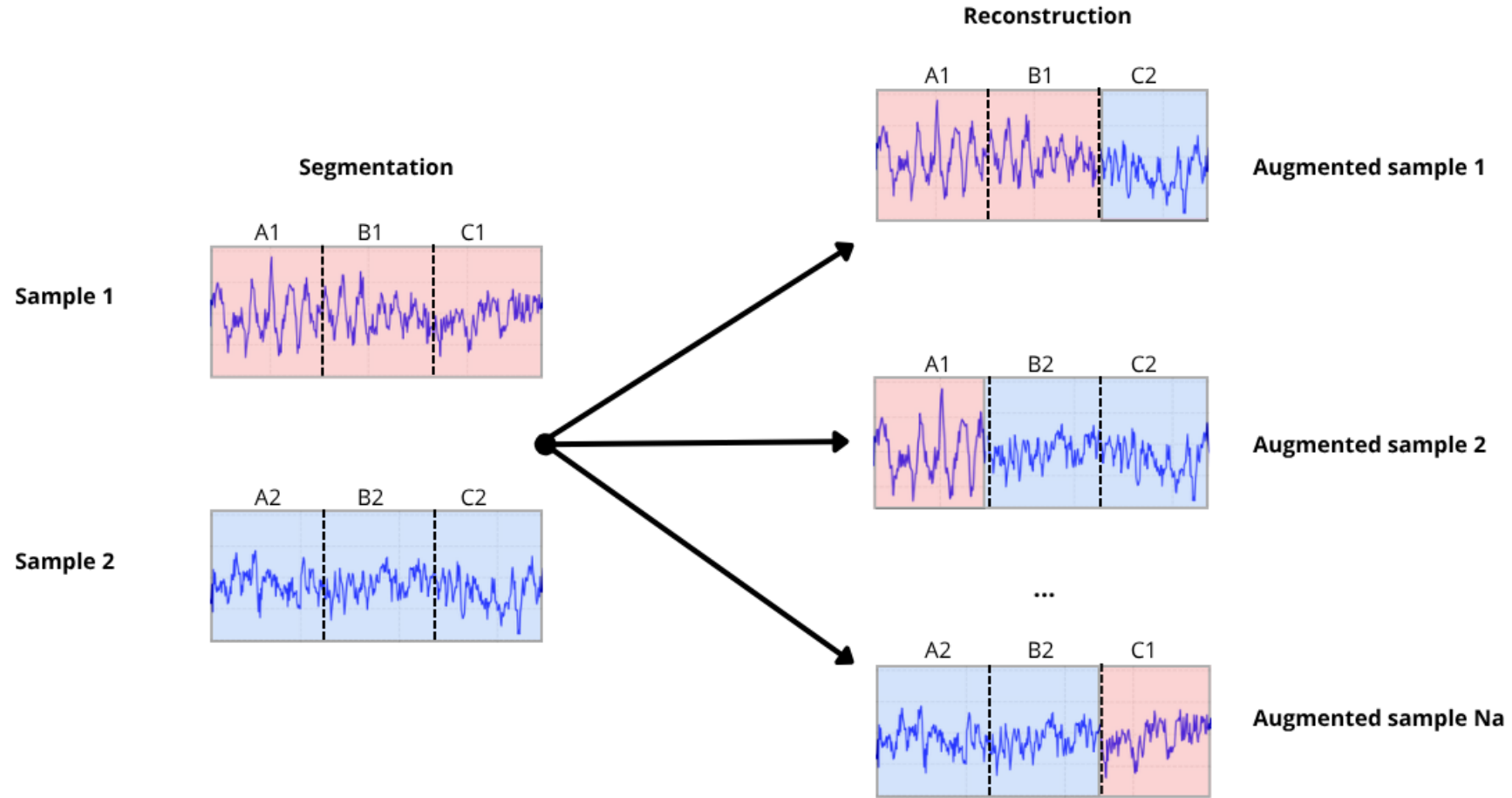


VIT lacks inductive bias



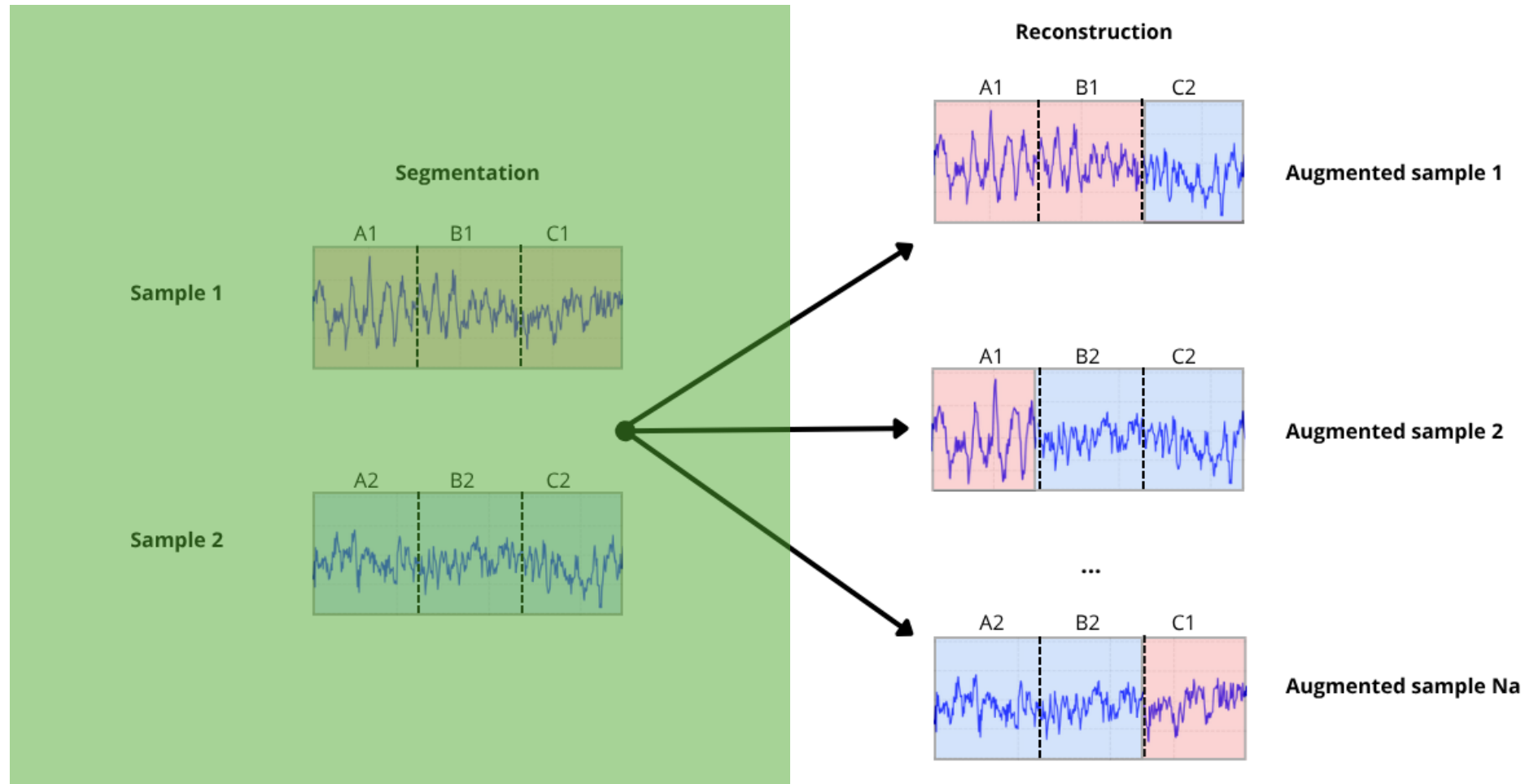
Requires more data

Intra-subject performance: S&R augmentation impact



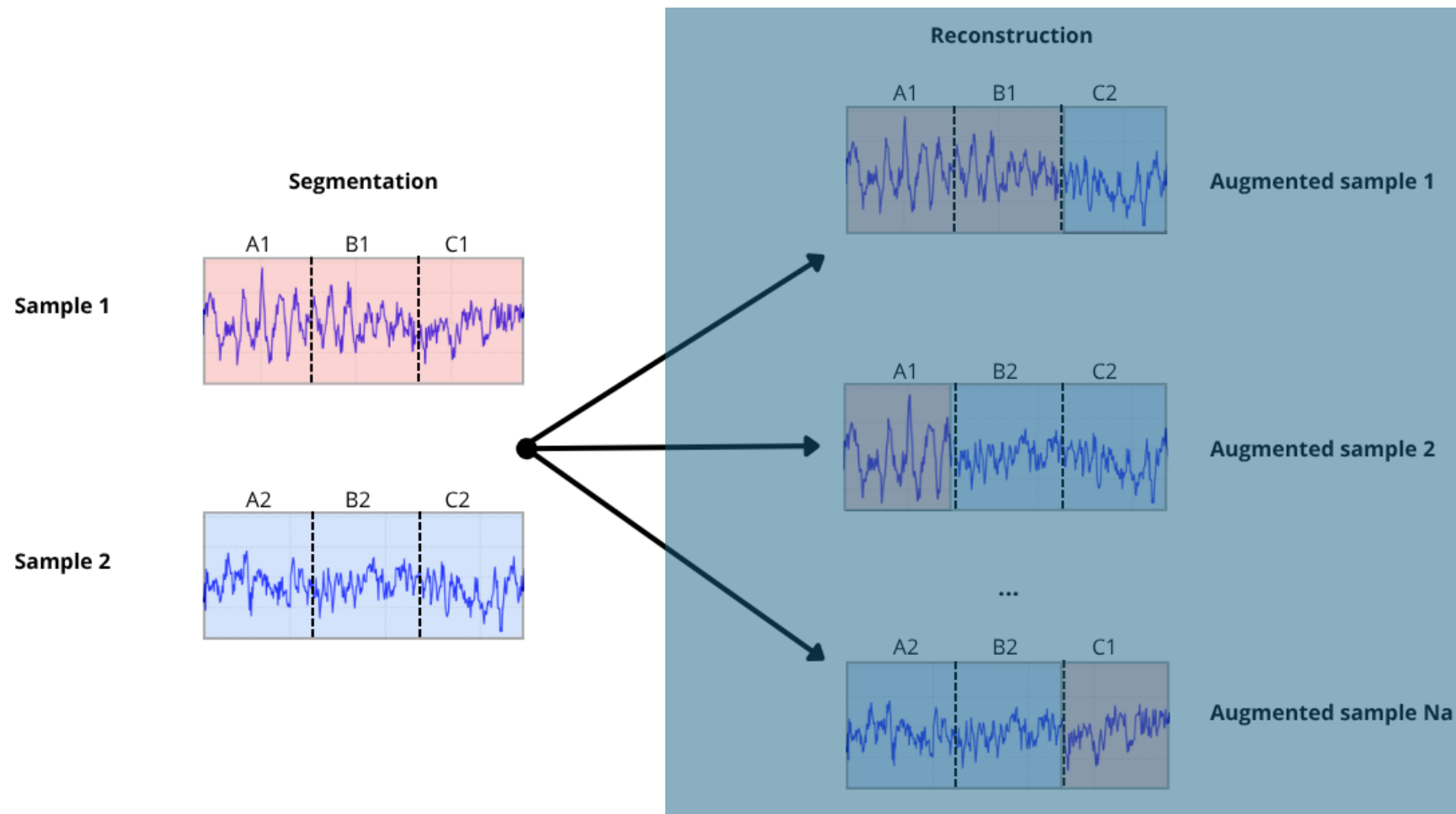
Intra-subject performance: S&R augmentation impact

Segmentation: each sample is divided into segments along the time axis



Intra-subject performance: S&R augmentation impact

Reconstruction: the segments are randomly recombined while maintaining temporal coherence



Intra-subject performance: S&R augmentation impact

Dataset: BCI competition IV 2a

No augmentation

Model	MeanAcc±std	MeanKappa±std
EEG-MVIT	59.45±14.53	0.46 ± 0.19
ENC-MVIT	60.00±15.76	0.46 ± 0.21
3BRANCH-VIT	61.23±14.72	0.48 ± 0.19
DEPTH-VIT	63.73±15.55	0.52 ± 0.20
LAMBDA-VIT	63.62±15.48	0.52 ± 0.20

With augmentation

Model	MeanAcc±std	MeanKappa±std
EEG-MVIT	61.00±13.63	0.48 ± 0.18
ENC-MVIT	59.91±13.93	0.46 ± 0.18
3BRANCH-VIT	63.39±15.11	0.51 ± 0.20
DEPTH-VIT	65.90±13.06	0.54 ± 0.17
LAMBDA-VIT	66.98±15.17	0.56 ± 0.20

**≈ +2% accuracy
improvement
on each model**

Intra-subject performance: S&R augmentation impact

Dataset: BCI competition IV 2b

No augmentation

Model	MeanAcc±std	MeanKappa±std
EEG-MVIT	77.08±14.95	0.54 ± 0.29
ENC-MVIT	76.70±15.76	0.56 ± 0.29
3BRANCH-VIT	79.00±13.62	0.58 ± 0.27
DEPTH-VIT	78.25±14.08	0.56 ± 0.28
LAMBDA-VIT	78.51±15.78	0.57 ± 0.31

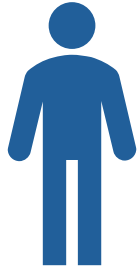
With augmentation

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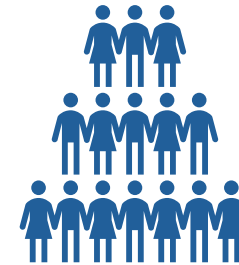
**≈ +1.5% accuracy
improvement on
each model**

Supervised pre-training

Intra-subject



Cross-subject



A supervised pre-training strategy is introduced, **combining both approaches**, when data from multiple patients is available, it is possible to:

Train a model on the training data of all patients



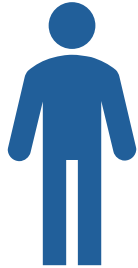
initialize the model weights



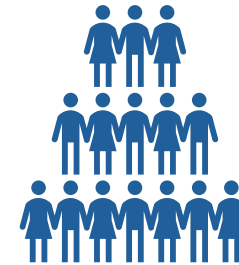
fine-tune it on the specific patient.

Supervised pre-training

Intra-subject



Cross-subject



Model	Dataset	MeanAcc±std	MeanKappa±std
EEG-MVIT (standard)	2a	61.00±13.63	0.48 ± 0.18
EEG-MVIT (supervised)		63.04±15.61	0.51 ± 0.20
EEG-MVIT (standard)	2b	78.15±13.50	0.56 ± 0.27
EEG-MVIT (supervised)		79.84±12.81	0.60 ± 0.25

**Another $\approx +2\%$
accuracy
improvement!**

First stage observations

Drawback of
the Inductive
bias

But



EEG-MVIT
has few CNN
layer

Data
scarcity

But



Unlabeled EEG signals
from any neurological
activity can be
leveraged.

Intra-subject
limits

But



Multi-subject
EEG signals
can be used

First stage conclusions

EEG-MVIT
has only one
CNN layer

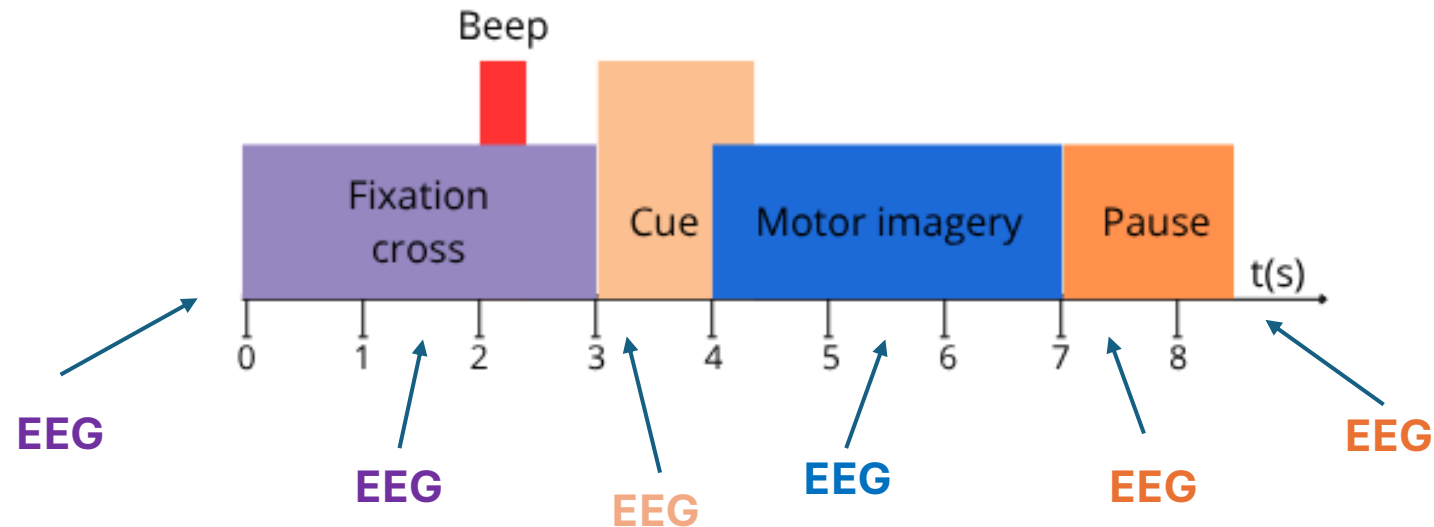
Unlabeled EEG signals
from any neurological
activity can be
leveraged.

Multi-subject
EEG signals
can be used



**Self-supervised pre-training to enhance
generalization**

Self-supervised pre-training to enhance generalization



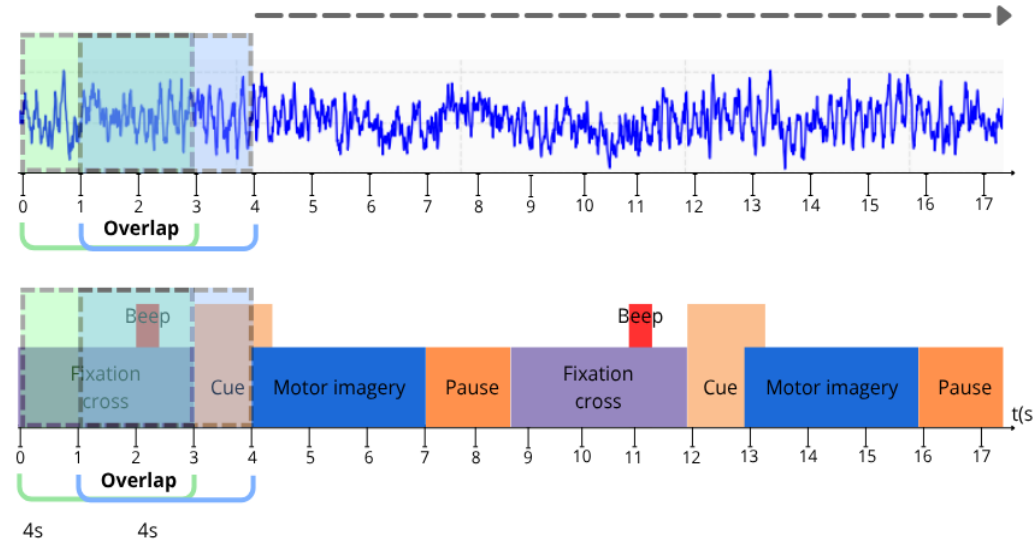
**EEG data are not limited
to motor imagery**

Self-supervised pre-training to enhance generalization

Data scarcity



Multi-subject EEG signals can be used

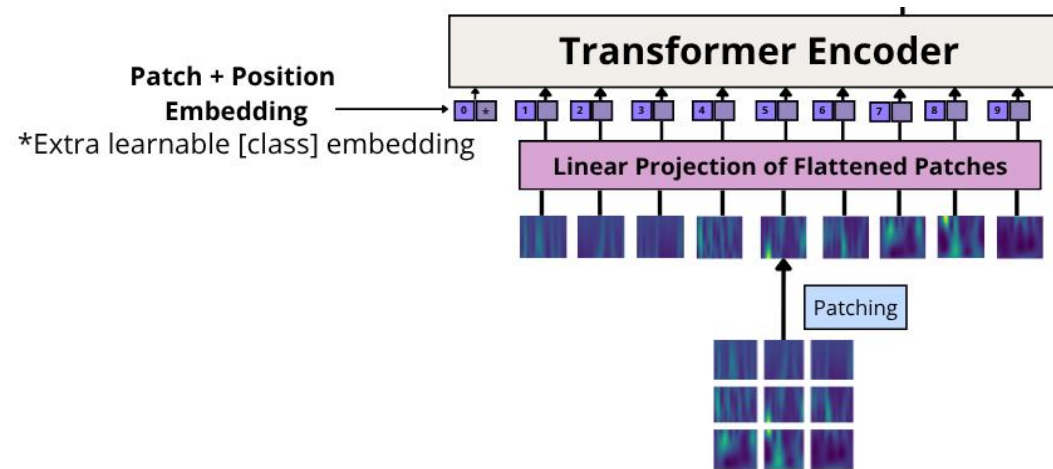


From 400 EEG samples per subject to about 6,000 per subject.

The full dataset consists of roughly 60,000 **unlabeled EEG** segments, obtained through **temporal window segmentation** with a stride of 1.

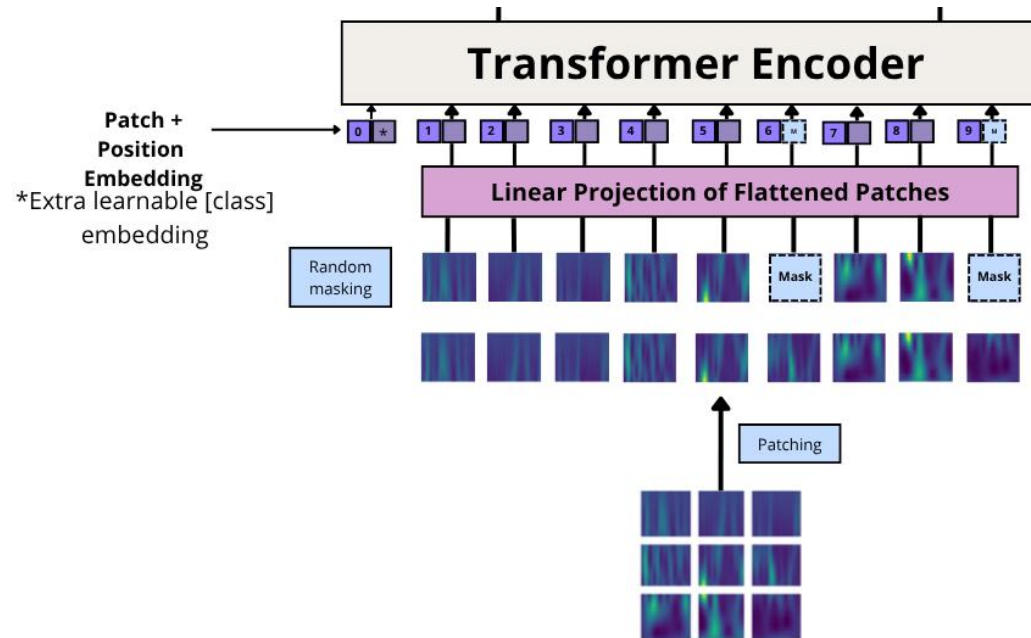
Self-supervised pre-training task

ViT relies on a **patch embedding layer** that divides the signal into patches before passing it to a Transformer Encoder



Self-supervised pre-training task

ViT relies on a **patch embedding layer** that divides the signal into patches before passing it to a Transformer Encoder

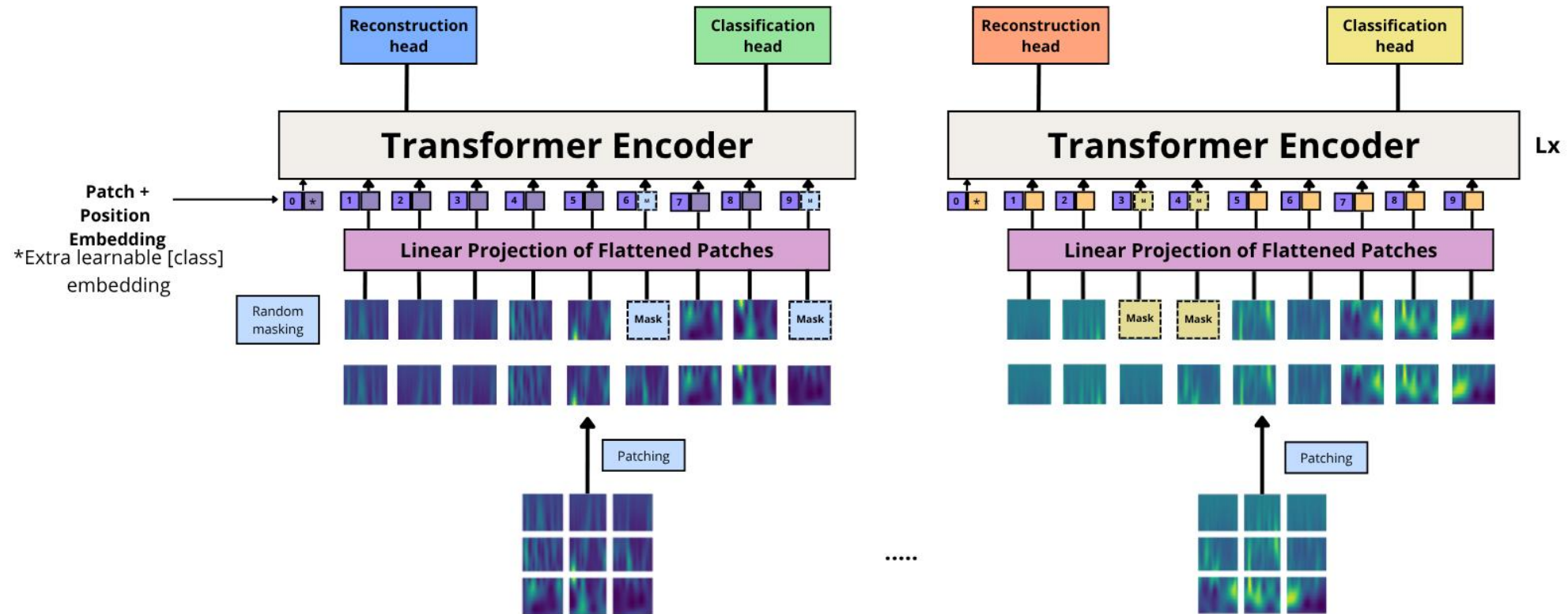


TASK: predicting masked patches

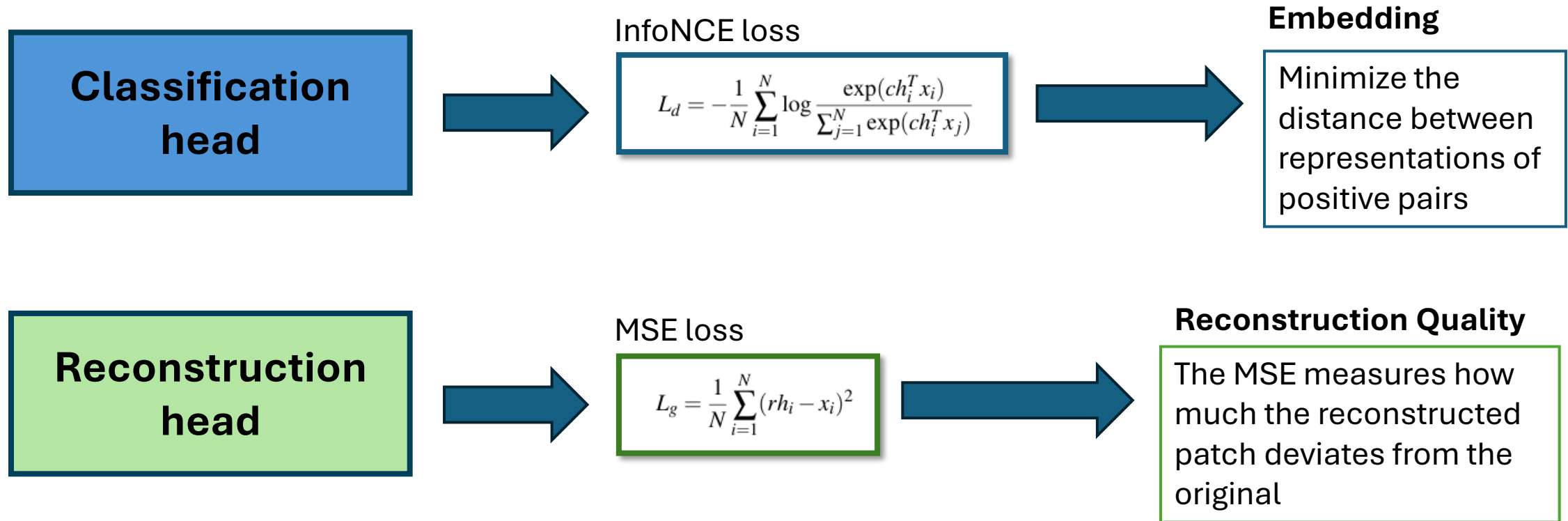


The model learns the underlying structure more deeply and can improve its interpretation of EEG data

Self-supervised pre-training task

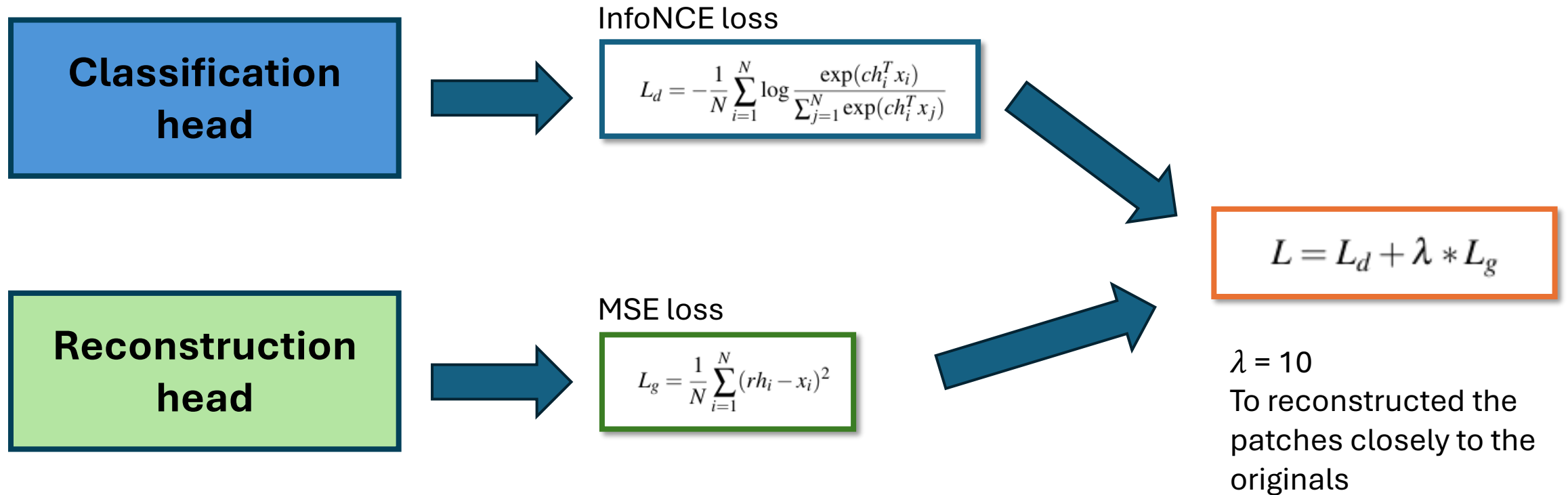


Self-supervised pre-training task



- $x_i \rightarrow$ removed patch i
- $ch_i \rightarrow$ Reconstructed patch i for classification head
- $rh_i \rightarrow$ Reconstructed patch i for reconstruction head
- $x_j \rightarrow$ removed patch j
- $N \rightarrow$ total masked patches

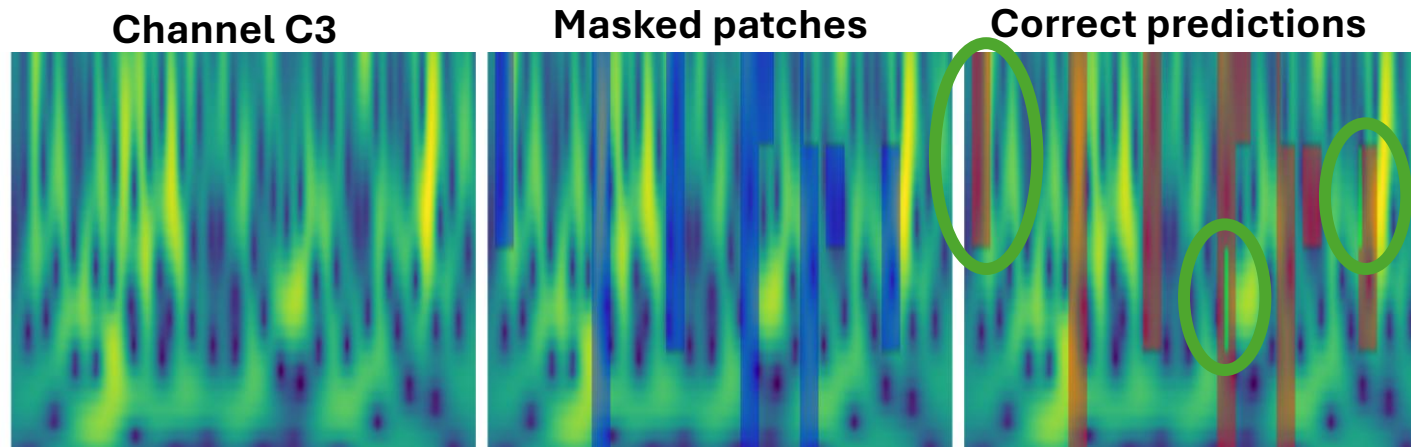
Self-supervised pre-training task



- $x_i \rightarrow$ removed patch i
- $ch_i \rightarrow$ Reconstructed patch i for classification head
- $rh_i \rightarrow$ Reconstructed patch i for reconstruction head
- $x_j \rightarrow$ removed patch j
- $N \rightarrow$ total masked patches

Self-supervised pre-training task: test

Start:
N = 100 random
masked patches



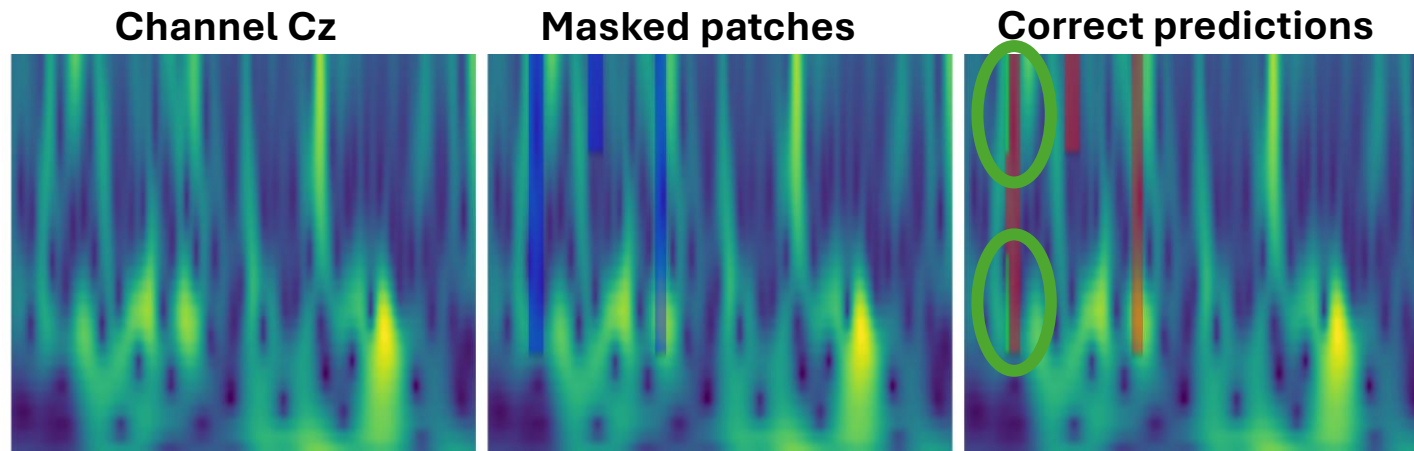
When the model
makes a correct guess,
the patch turns from
red to **green**.

Few predicted patches



Self-supervised pre-training task: test

Start:
N = 25 random
masked patches



When the model
makes a correct guess,
the patch turns from
red to **green**.

After 80 epochs ...

Few predicted patches



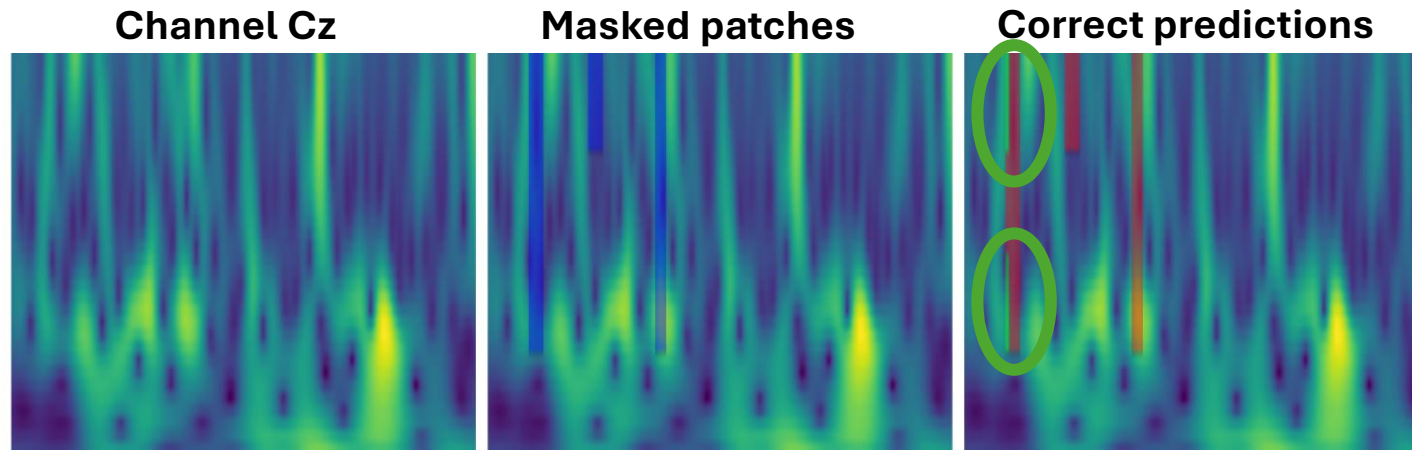
Let's make the task easier by decreasing
the total number of masked patches from
N=100 to N=25



NOTHING CHANGED !

Self-supervised pre-training task: test

Start:
N = 25 random
masked patches



When the model
makes a correct guess,
the patch turns from
red to **green**.

After 80 epochs ...

Few predicted patches



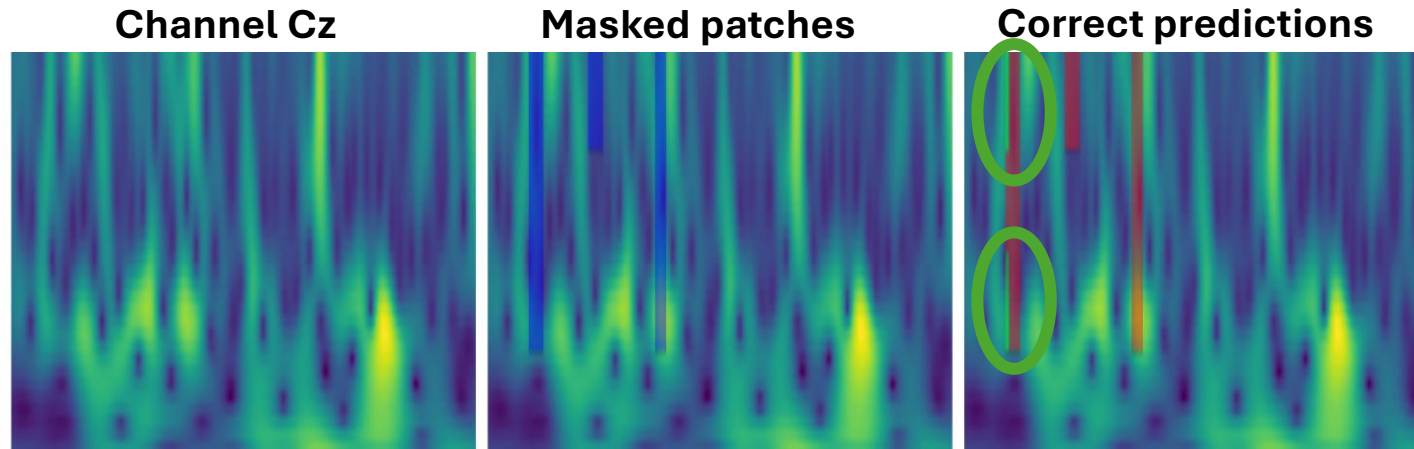
Let's make the task easier by decreasing
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NOTHING CHANGED !

Self-supervised pre-training task: test

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After 80 epochs ...

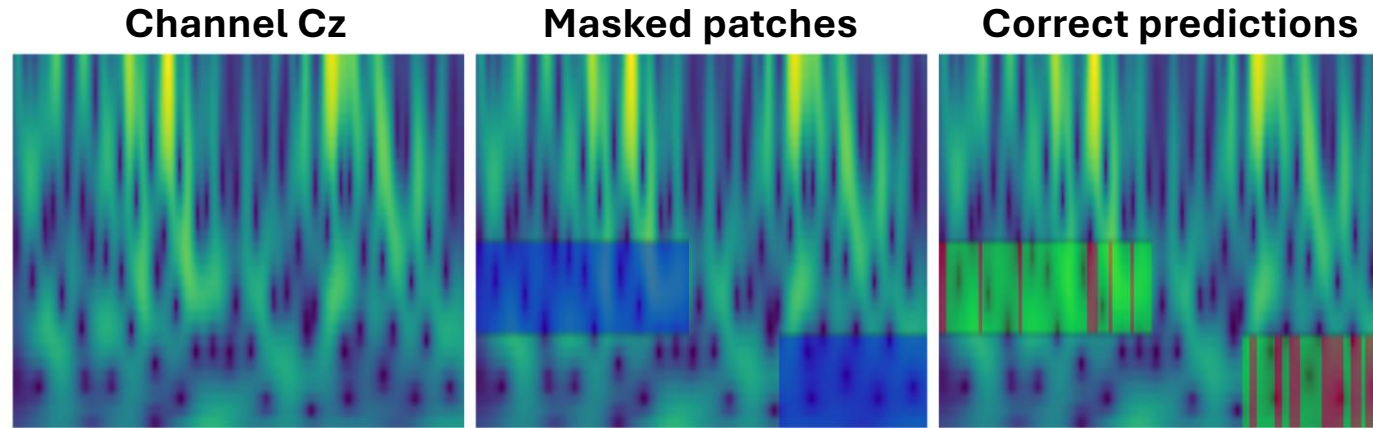
Few predicted patches



Let's make the task easier by masking
N = 100 adjacent patches.

Self-supervised pre-training task: test

Start:
N = 100 adjacent
masked patches



When the model
makes a correct guess,
the patch turns from
red to **green**.

Few predicted patches



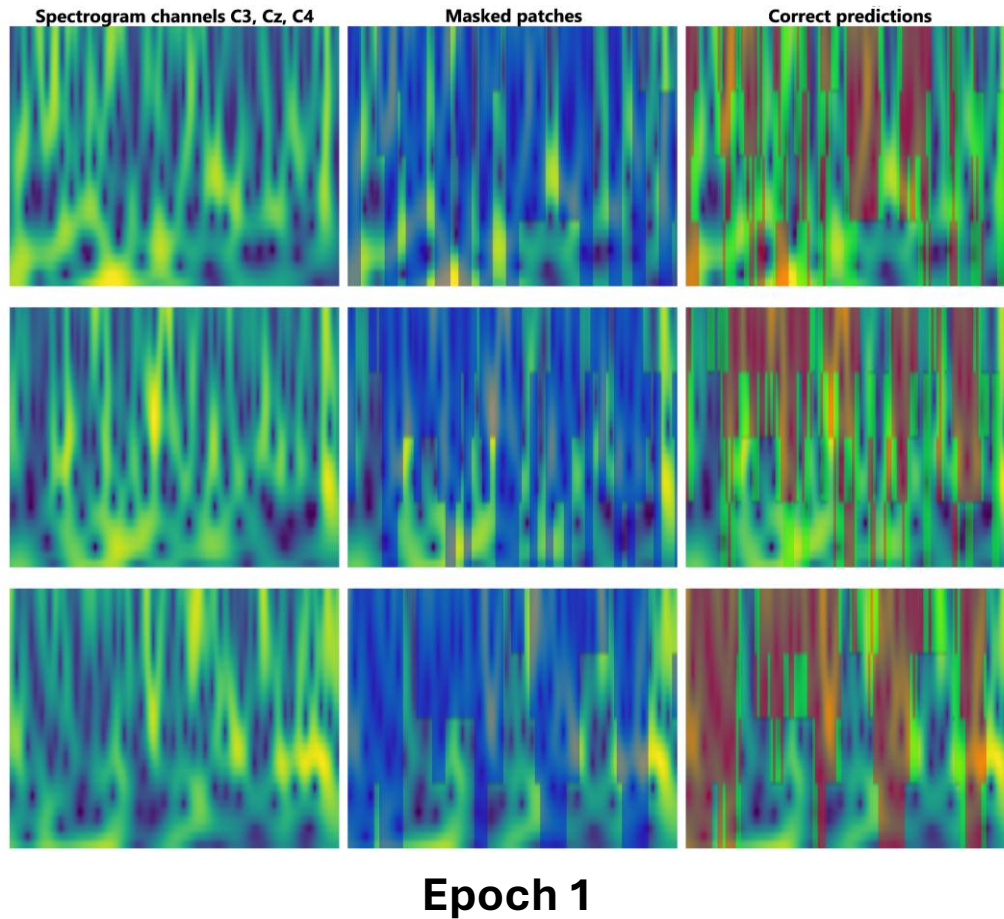
Let's make the task easier by masking
N = 100 adjacent patches.



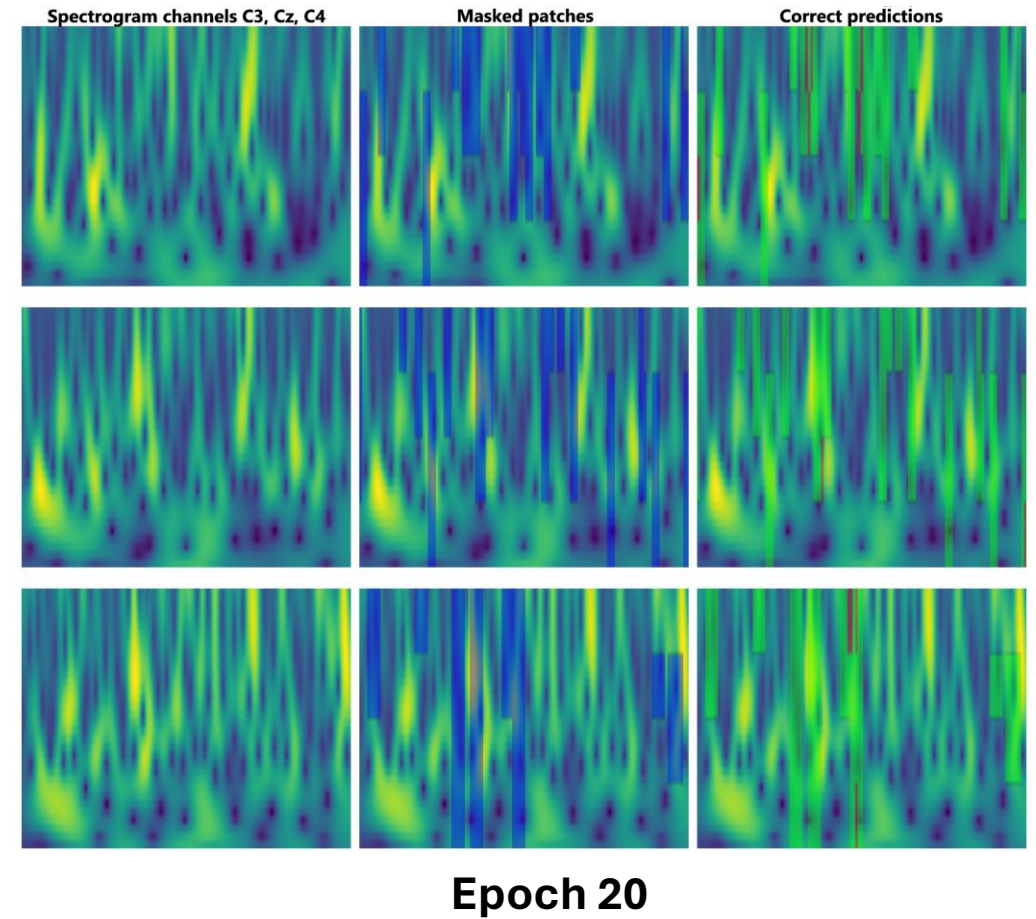
It worked!
It could be the
basis for the more
complex tasks.

Curriculum Self-supervised pre-training: medium task

N=100

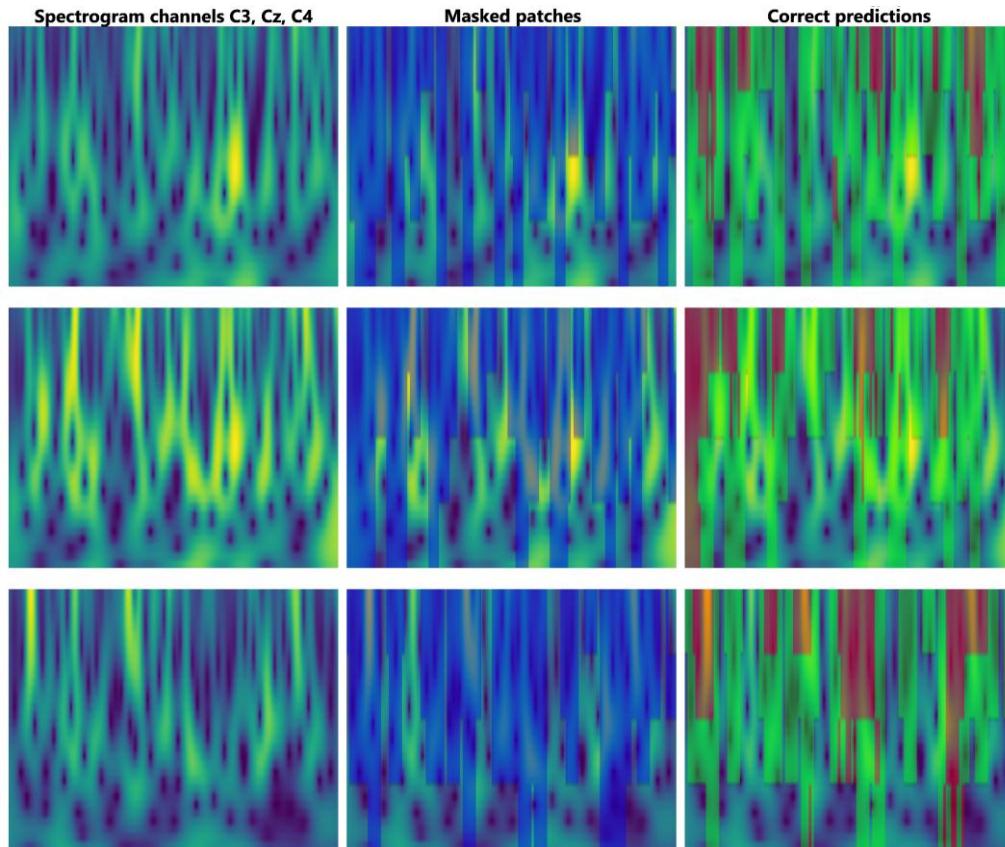


...



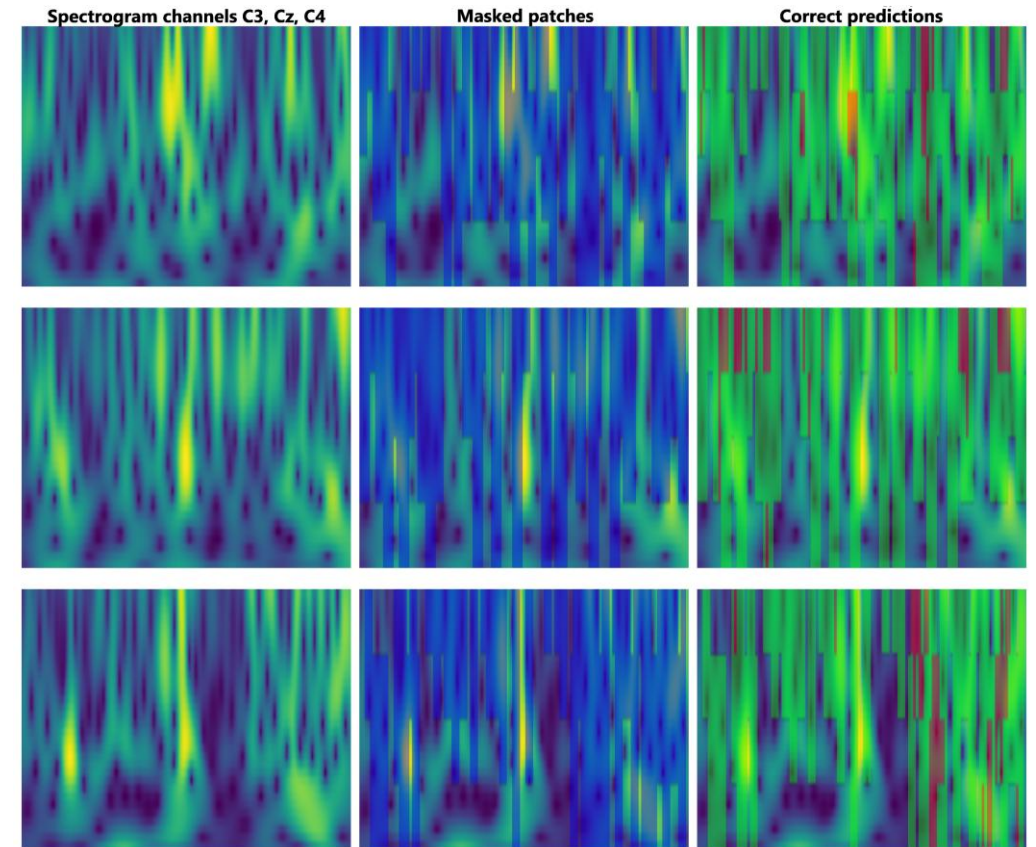
Curriculum Self-supervised pre-training: hard task

N=300



Epoch 1

...

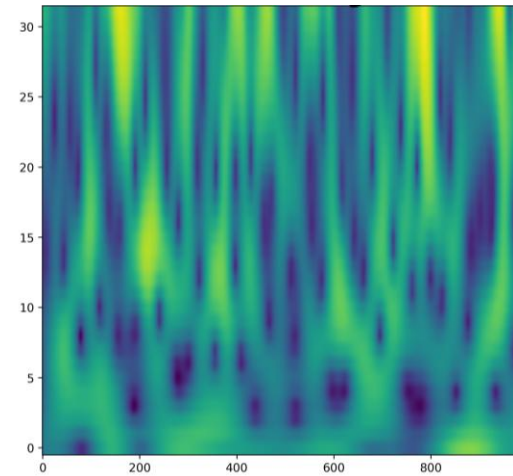


Epoch 100

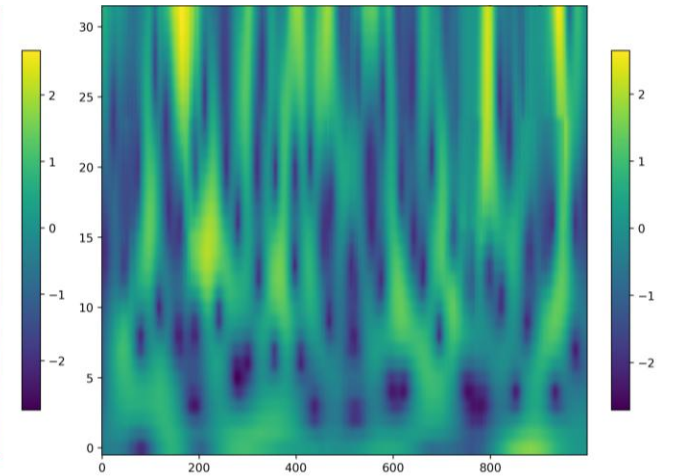
Curriculum Self-supervised pre-training pipeline

1. Apply **continuous masking** to 100 patches
2. Apply **random masking** with 100, and 300 patches
3. **Fine-tune** the model

Channel Cz - Original



Reconstructed (N = 250)



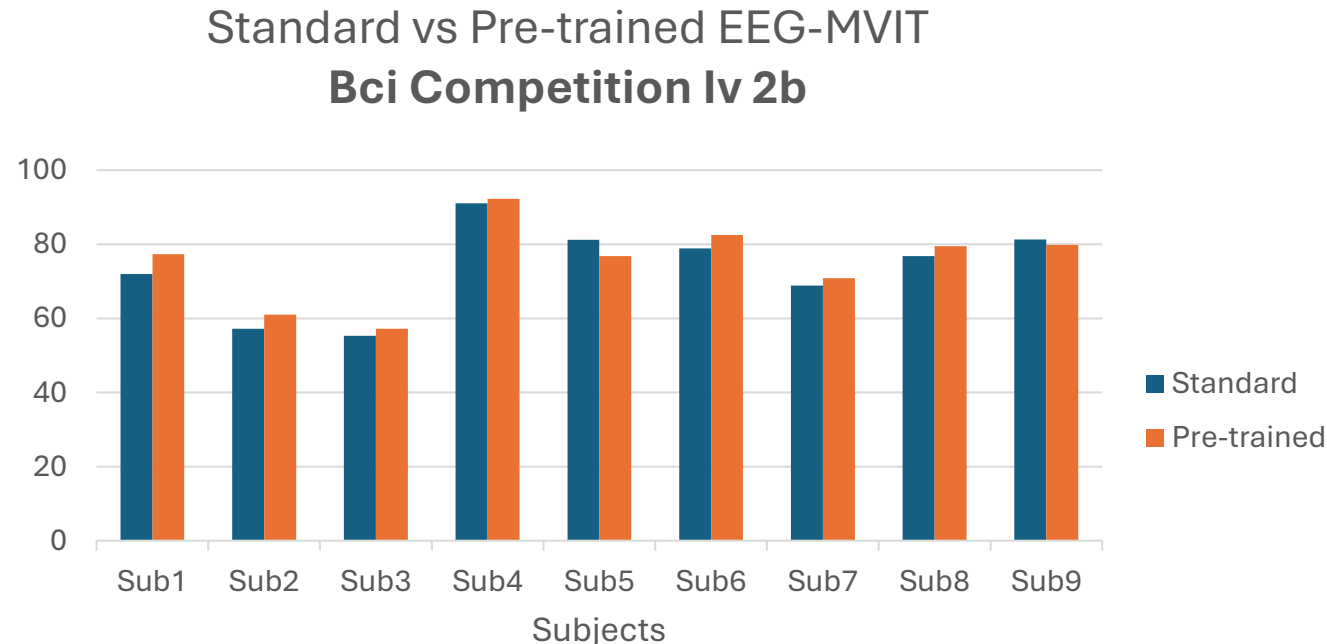
With (8x8) patches, the model is able to reconstruct 60% of the spectrogram

Curriculum Self-supervised pre-training EEG-MVIT cross-subject results on 2b

Model	MeanAcc $\pm$ std	MeanKappa $\pm$ std
Standard	73.62 $\pm$ 11.01	0.47 $\pm$ 0.22
Self Supervised (**)	75.26$\pm$10.20	0.51 $\pm$ 0.20

*A p-value Wilcoxon test was performed: $p < 0.01$ (**) indicates that the performance improvement is statistically significant.*

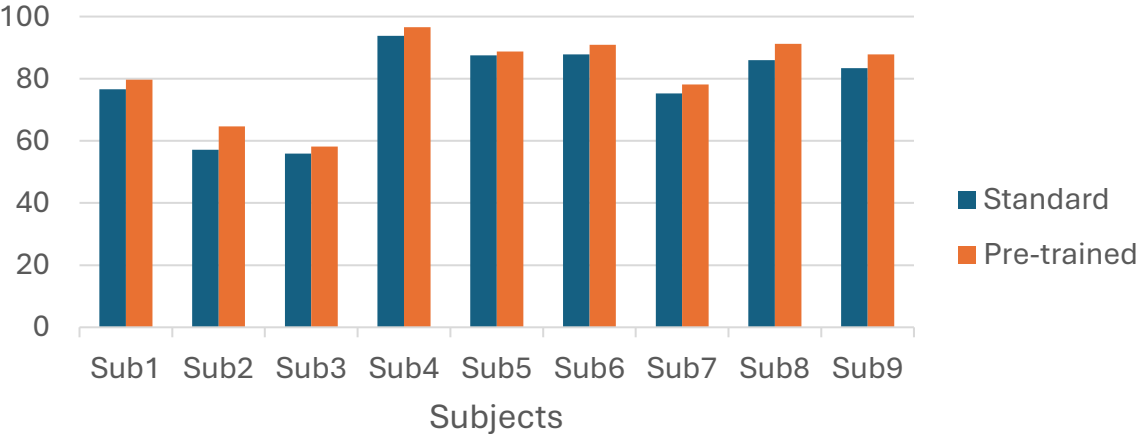
**$\approx +2\%$
accuracy
improvement
thanks to
pretrain!**



Curriculum Self-supervised pre-training EEG-MVIT intra-subject results

Model	MeanAcc±std	MeanKappa±std
No augmentation (**)	77.08±14.95	0.54 ± 0.29
Standard	78.15±13.50	0.56 ± 0.27
Supervised (**)	79.84±12.81	0.60±0.25
Supervised + Self Supervised (**)	81.77±12.98	0.64 ± 0.25

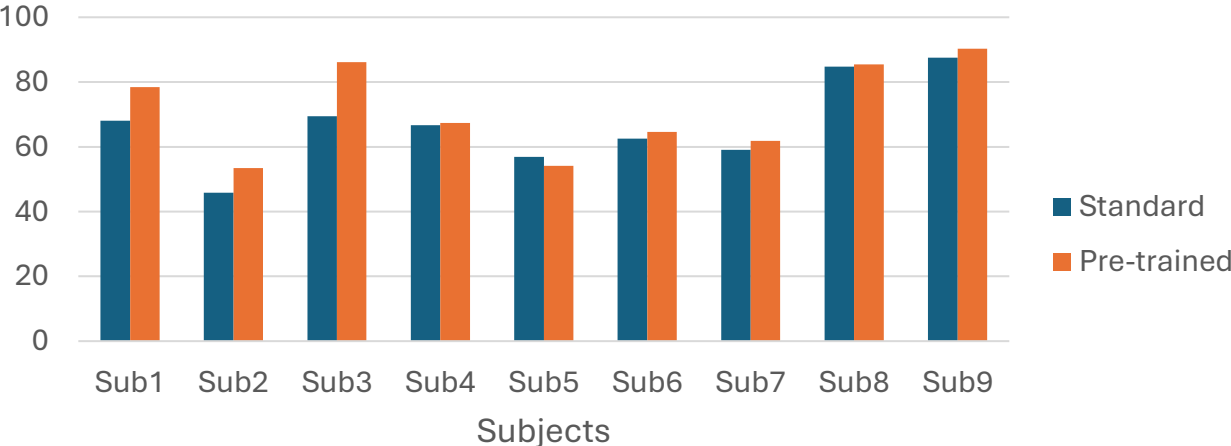
Standard vs Pre-trained EEG-MVIT
Bci Competition Iv 2b



Model	MeanAcc±std	MeanKappa±std
Standard	66.74±13.11	0.33 ± 0.13
Supervised(*)	68.83±13.5	0.38 ± 0.27
Supervised + Self Supervised (**)	71.30±0.14	0.42 ± 0.28

+5%

Standard vs Pre-trained EEG-MVIT
Bci Competition Iv 2a 2 class



Conclusions



Conclusions

Managed by Curriculum Self
Supervised Pre-Training on
EEG-MVIT network

Utilization of the **ViT** and
the **patch-embedding
layer**

Ability to leverage **EEG**
data from **multiple
fields**

Ability to leverage **EEG**
from **multiple subjects**

More **general** features



≈ +2% cross-subject
≈ +4% intra-subject
Improvement on 2b
thanks to pre-train!

Conclusions

Managed by Curriculum Self
Supervised Pre-Training on
EEG-MVIT network

Utilization of the **ViT** and
the **patch-embedding
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Ability to leverage **EEG**
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More **general** features



**New motor imagery
framework!**

Future improvements

Temple University Hospital (TUH) dataset [9] has **69,652 recordings** divided between healthy and epileptic patients



Each file contains over 10,000 samples



**Approximately
600,000
spectrograms!**

Pre-training on an even bigger dataset

Future improvements

Temple University Hospital (TUH) dataset [9] has **69,652 recordings** divided between healthy and epileptic patients



Each file contains over 10,000 samples



**Approximately
600,000
spectrograms!**

Pre-training on an even bigger dataset

A few-epoch test showed that convergence takes much longer...

Starting from the self supervised pre-trained model presented an increase in convergence speed!

EEG-MViT web interface

Deploy ⋮

 **EEG Classification with MultiChannelViT**

Select the dataset first, then upload a .gdf file, view raw EEG, preprocess, and run inference.

 Reset to dataset selection

Select Dataset ⇄

Choose your dataset

☐ 2a
☒ 2b

Confirm Dataset Selection

EEG Classification with MultiChannelViT

Select the dataset first, then upload a .gdf file, view raw EEG, preprocess, and run inference.

 Reset to dataset selection

Select Dataset ↔

Choose your dataset

☐ 2a

☒ 2b

Confirm Dataset Selection

Thank you for your attention.



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